

## **Supplementary Material**

### **Evolutionary Constraint of Recurrent Human TP53 Cancer Mutation Hotspots Across Mammalian Species**

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This supplementary document organizes the supplied figures, tables, references, source-data descriptions,  
audit records and reproducibility information without modifying scientific results.

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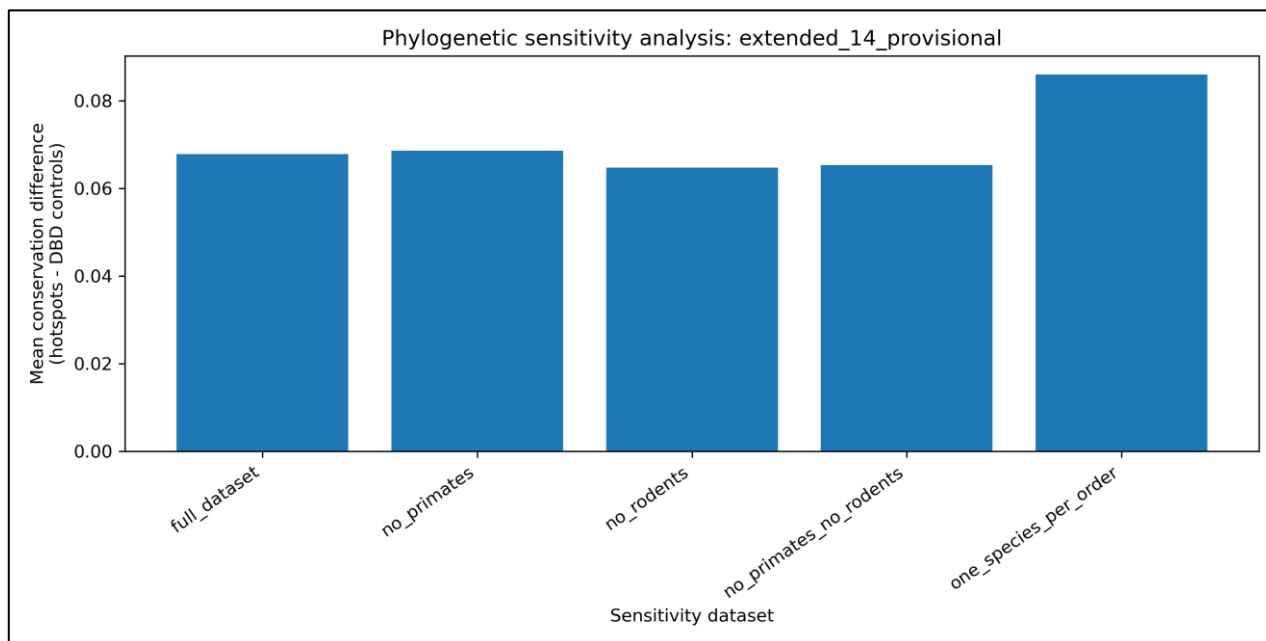
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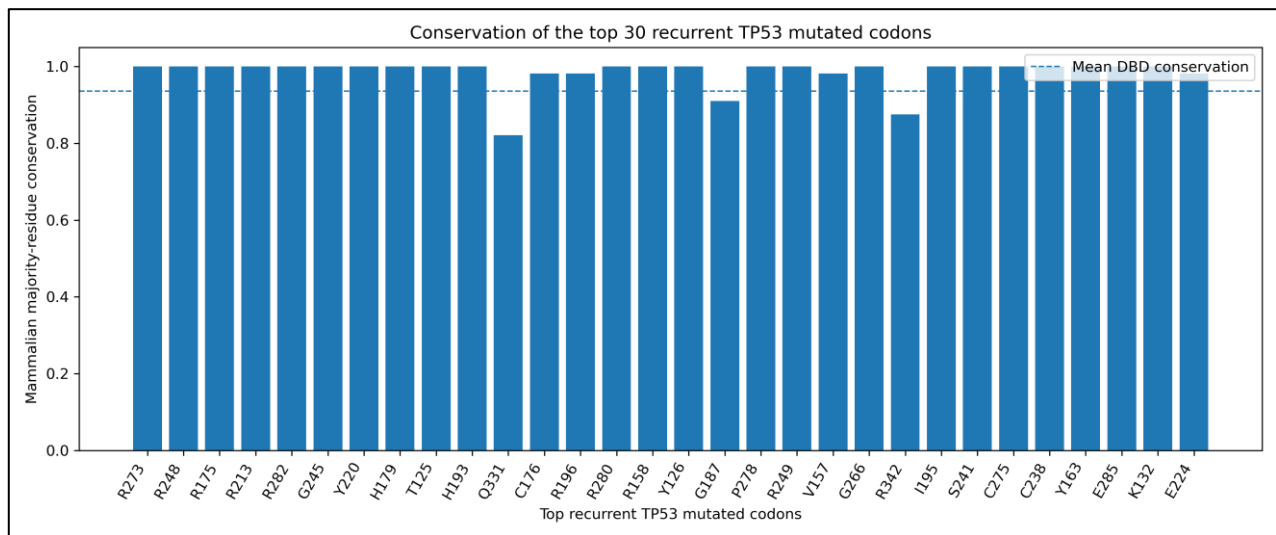
### Supplementary Figures

Figure S1



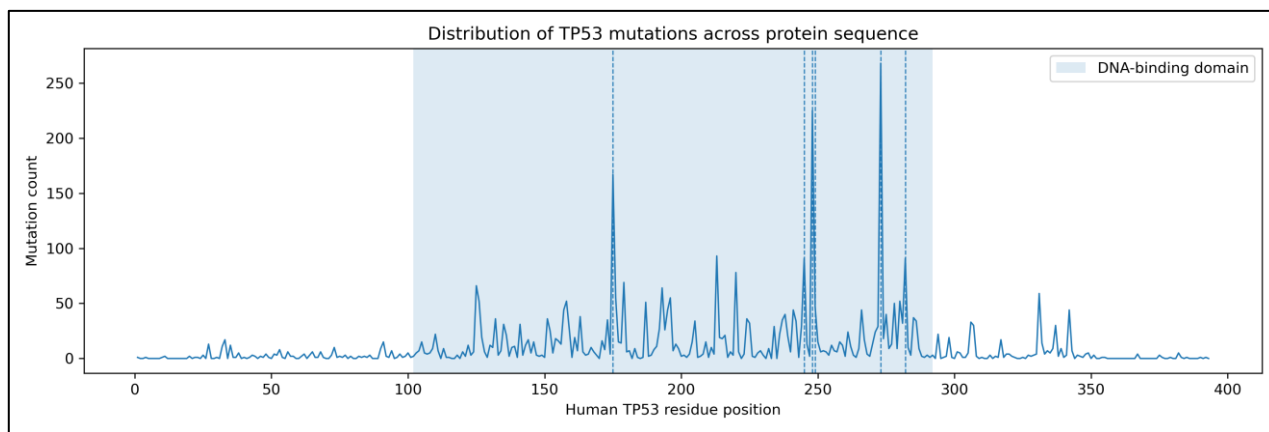
**Figure S1.** Extended phylogenetic sensitivity analysis of TP53 hotspot conservation. The figure summarizes conservation comparisons across reduced mammalian datasets and shows whether the hotspot-conservation signal remains stable after lineage-exclusion and downsampling strategies.

**Figure S2**



**Figure S2.** Conservation of the top 30 recurrently mutated TP53 codons. The figure displays residue-level mammalian conservation for highly recurrent human TP53 cancer mutation codons, supporting comparison between mutation recurrence and evolutionary constraint.

Figure S3



**Figure S3.** Distribution of TP53 mutation recurrence across human codons. The figure summarizes codon-level mutation-frequency patterns from cBioPortal/TCGA PanCancer data and highlights recurrent mutation positions used for integration with mammalian conservation metrics.

## Supplementary Tables

All supplementary tables were rendered as editable tables from the supplied CSV files. No table rows were truncated.

**Table S1.** TP53 sequence accession audit.

| accession    | species                    | common name                    | order          | sequence length | header                                                                          | audit flag |
|--------------|----------------------------|--------------------------------|----------------|-----------------|---------------------------------------------------------------------------------|------------|
| NP_001394193 | Homo sapiens               | Human                          | Primates       | 393             | NP_001394193 Homo_sapiens Human Primates                                        | PASS       |
| XP_001172077 | Pan troglodytes            | Chimpanzee                     | Primates       | 393             | XP_001172077 Pan_troglodytes Chimpanzee Primates                                | PASS       |
| XP_063556659 | Gorilla gorilla            | Gorilla                        | Primates       | 408             | XP_063556659 Gorilla_gorilla Gorilla Primates                                   | PASS       |
| XP_002827020 | Pongo abelii               | Orangutan                      | Primates       | 393             | XP_002827020 Pongo_abelii Orangutan Primates                                    | PASS       |
| NP_001040616 | Macaca mulatta             | Rhesus_macaque                 | Primates       | 393             | NP_001040616 Macaca_mulatta Rhesus_macaque Primates                             | PASS       |
| XP_003912321 | Papio anubis               | Baboon                         | Primates       | 393             | XP_003912321 Papio_anubis Baboon Primates                                       | PASS       |
| XP_002747994 | Callithrix jacchus         | Marmoset                       | Primates       | 395             | XP_002747994 Callithrix_jacchus Marmoset Primates                               | PASS       |
| XP_012631512 | Microcebus murinus         | Mouse_lemur                    | Primates       | 385             | XP_012631512 Microcebus_murinus Mouse_lemur Primates                            | PASS       |
| NP_001120705 | Mus musculus               | Mouse                          | Rodentia       | 381             | NP_001120705 Mus_musculus Mouse Rodentia                                        | PASS       |
| NP_001416925 | Rattus norvegicus          | Rat                            | Rodentia       | 382             | NP_001416925 Rattus_norvegicus Rat Rodentia                                     | PASS       |
| NP_001230905 | Cricetulus griseus         | Chinese_hamster                | Rodentia       | 393             | NP_001230905 Cricetulus_griseus Chinese_hamster Rodentia                        | PASS       |
| NP_001166211 | Cavia porcellus            | Guinea_pig                     | Rodentia       | 391             | NP_001166211 Cavia_porcellus Guinea_pig Rodentia                                | PASS       |
| XP_040125303 | Ictidomys tridecemlineatus | Thirteen-lined_ground_squirrel | Rodentia       | 391             | XP_040125303 Ictidomys_tridecemlineatus Thirteen-lined_ground_squirrel Rodentia | PASS       |
| NP_001297199 | Heterocephalus glaber      | Naked_mole-rat                 | Rodentia       | 391             | NP_001297199 Heterocephalus_glaber Naked_mole-rat Rodentia                      | PASS       |
| XP_005349834 | Microtus ochrogaster       | Prairie_vole                   | Rodentia       | 390             | XP_005349834 Microtus_ochrogaster Prairie_vole Rodentia                         | PASS       |
| XP_012868945 | Dipodomys ordii            | Kangaroo_rat                   | Rodentia       | 398             | XP_012868945 Dipodomys_ordii Kangaroo_rat Rodentia                              | PASS       |
| NP_001376147 | Canis lupus familiaris     | Dog                            | Carnivora      | 381             | NP_001376147 Canis_lupus_familiaris Dog Carnivora                               | PASS       |
| NP_001009294 | Felis catus                | Cat                            | Carnivora      | 386             | NP_001009294 Felis_catus Cat Carnivora                                          | PASS       |
| XP_057167334 | Ursus arctos               | Brown_bear                     | Carnivora      | 378             | XP_057167334 Ursus_arctos Brown_bear Carnivora                                  | PASS       |
| XP_004760313 | Mustela putorius furo      | Ferret                         | Carnivora      | 375             | XP_004760313 Mustela_putorius_furo Ferret Carnivora                             | PASS       |
| XP_019660679 | Ailuropoda melanoleuca     | Giant_panda                    | Carnivora      | 381             | XP_019660679 Ailuropoda_melanoleuca Giant_panda Carnivora                       | PASS       |
| XP_042822088 | Panthera tigris            | Tiger                          | Carnivora      | 386             | XP_042822088 Panthera_tigris Tiger Carnivora                                    | PASS       |
| XP_025861192 | Vulpes vulpes              | Red_fox                        | Carnivora      | 381             | XP_025861192 Vulpes_vulpes Red_fox Carnivora                                    | PASS       |
| XP_027481717 | Zalophus californianus     | California_sea_lion            | Carnivora      | 381             | XP_027481717 Zalophus_californianus California_sea_lion Carnivora               | PASS       |
| NP_776626    | Bos taurus                 | Cattle                         | Artiodactyla   | 386             | NP_776626 Bos_taurus Cattle Artiodactyla                                        | PASS       |
| NP_998989    | Sus scrofa                 | Pig                            | Artiodactyla   | 386             | NP_998989 Sus_scrofa Pig Artiodactyla                                           | PASS       |
| NP_001009403 | Ovis aries                 | Sheep                          | Artiodactyla   | 382             | NP_001009403 Ovis_aries Sheep Artiodactyla                                      | PASS       |
| XP_017920082 | Capra hircus               | Goat                           | Artiodactyla   | 382             | XP_017920082 Capra_hircus Goat Artiodactyla                                     | PASS       |
| XP_010994816 | Camelus dromedarius        | Dromedary_camel                | Artiodactyla   | 381             | XP_010994816 Camelus_dromedarius Dromedary_camel Artiodactyla                   | PASS       |
| XP_006218839 | Vicugna pacos              | Alpaca                         | Artiodactyla   | 381             | XP_006218839 Vicugna_pacos Alpaca Artiodactyla                                  | PASS       |
| XP_057572051 | Hippopotamus amphibius     | Hippopotamus                   | Artiodactyla   | 386             | XP_057572051 Hippopotamus_amphibius Hippopotamus Artiodactyla                   | PASS       |
| NP_001189334 | Equus caballus             | Horse                          | Perissodactyla | 381             | NP_001189334 Equus_caballus Horse Perissodactyla                                | PASS       |
| XP_014717043 | Equus asinus               | Donkey                         | Perissodactyla | 381             | XP_014717043 Equus_asinus Donkey Perissodactyla                                 | PASS       |
| XP_014645038 | Ceratotherium simum        | White_rhinoceros               | Perissodactyla | 397             | XP_014645038 Ceratotherium_simum White_rhinoceros Perissodactyla                | PASS       |
| XP_003416950 | Loxodonta africana         | African_elephant               | Proboscidea    | 390             | XP_003416950 Loxodonta_africana African_elephant Proboscidea                    | PASS       |

| accession    | species                   | common name           | order           | sequence length | header                                                                  | audit flag |
|--------------|---------------------------|-----------------------|-----------------|-----------------|-------------------------------------------------------------------------|------------|
| XP_049714739 | Elephas maximus           | Asian_elephant        | Proboscidea     | 390             | XP_049714739 Elephas_maximus Asian_elephant Proboscidea                 | PASS       |
| XP_036692565 | Balaenoptera musculus     | Blue_whale            | Cetacea         | 387             | XP_036692565 Balaenoptera_musculus Blue_whale Cetacea                   | PASS       |
| XP_073653870 | Tursiops truncatus        | Bottlenose_dolphin    | Cetacea         | 393             | XP_073653870 Tursiops_truncatus Bottlenose_dolphin Cetacea              | PASS       |
| XP_033278862 | Orcinus orca              | Orca                  | Cetacea         | 400             | XP_033278862 Orcinus_orca Orca Cetacea                                  | PASS       |
| XP_028355209 | Physeter catodon          | Sperm_whale           | Cetacea         | 387             | XP_028355209 Physeter_catodon Sperm_whale Cetacea                       | PASS       |
| XP_022451328 | Delphinapterus leucas     | Beluga_whale          | Cetacea         | 387             | XP_022451328 Delphinapterus_leucas Beluga_whale Cetacea                 | PASS       |
| XP_006102640 | Myotis lucifugus          | Little_brown_bat      | Chiroptera      | 390             | XP_006102640 Myotis_lucifugus Little_brown_bat Chiroptera               | PASS       |
| XP_011357167 | Pteropus vampyrus         | Large_flying_fox      | Chiroptera      | 378             | XP_011357167 Pteropus_vampyrus Large_flying_fox Chiroptera              | PASS       |
| XP_036078506 | Rousettus aegyptiacus     | Egyptian_fruit_bat    | Chiroptera      | 383             | XP_036078506 Rousettus_aegyptiacus Egyptian_fruit_bat Chiroptera        | PASS       |
| XP_008151723 | Eptesicus fuscus          | Big_brown_bat         | Chiroptera      | 388             | XP_008151723 Eptesicus_fuscus Big_brown_bat Chiroptera                  | PASS       |
| XP_032947086 | Rhinolophus ferrumequinum | Greater_horseshoe_bat | Chiroptera      | 380             | XP_032947086 Rhinolophus_ferrumequinum Greater_horseshoe_bat Chiroptera | PASS       |
| XP_056673571 | Monodelphis domestica     | Opossum               | Didelphimorphia | 361             | XP_056673571 Monodelphis_domestica Opossum Didelphimorphia              | PASS       |
| XP_031821305 | Sarcophilus harrisii      | Tasmanian_devil       | Dasyuromorphia  | 371             | XP_031821305 Sarcophilus_harrisii Tasmanian_devil Dasyuromorphia        | PASS       |
| XP_072497274 | Macropus eugenii          | Tammar_wallaby        | Diprotodontia   | 371             | XP_072497274 Macropus_eugenii Tammar_wallaby Diprotodontia              | PASS       |
| XP_020822127 | Phascolarctos cinereus    | Koala                 | Diprotodontia   | 371             | XP_020822127 Phascolarctos_cinereus Koala Diprotodontia                 | PASS       |
| XP_060060302 | Erinaceus europaeus       | European_hedgehog     | Eulipotyphla    | 374             | XP_060060302 Erinaceus_europaeus European_hedgehog Eulipotyphla         | PASS       |
| XP_054988705 | Sorex araneus             | Common_shrew          | Eulipotyphla    | 383             | XP_054988705 Sorex_araneus Common_shrew Eulipotyphla                    | PASS       |
| NP_001075873 | Oryctolagus cuniculus     | Rabbit                | Lagomorpha      | 391             | NP_001075873 Oryctolagus_cuniculus Rabbit Lagomorpha                    | PASS       |
| XP_004594840 | Ochotona princeps         | Pika                  | Lagomorpha      | 385             | XP_004594840 Ochotona_princeps Pika Lagomorpha                          | PASS       |
| XP_023447566 | Dasypus novemcinctus      | Nine-banded_armadillo | Cingulata       | 384             | XP_023447566 Dasypus_novemcinctus Nine-banded_armadillo Cingulata       | PASS       |
| XP_023586869 | Trichechus manatus        | Manatee               | Sirenia         | 392             | XP_023586869 Trichechus_manatus Manatee Sirenia                         | PASS       |

**Table S2.** Permutation hotspot statistical analysis.

| comparison                              | n query | observed mean conservation | null mean conservation | null 2.5 percentile | null 97.5 percentile | empirical p one sided |
|-----------------------------------------|---------|----------------------------|------------------------|---------------------|----------------------|-----------------------|
| canonical 6 vs DBD permutation          | 6       | 1                          | 0.935845368            | 0.8125              | 0.99702381           | 0.023589764           |
| top10 mutated codons vs DBD permutation | 10      | 1                          | 0.935881298            | 0.846428571         | 0.991071429          | 0.001819982           |
| top20 mutated codons vs DBD permutation | 19      | 0.992481203                | 0.935910678            | 0.876879699         | 0.979323308          | 0.001389986           |
| top30 mutated codons vs DBD permutation | 28      | 0.994260204                | 0.935896467            | 0.889587199         | 0.973133117          | 3.00E-05              |

**Table S3.** Residue-level conservation metrics, part 1 of 2: residue annotation and alignment coverage fields.

| human position | human residue | domain                 | is canonical hotspot | total species | non gap count | gap count | gap fraction |
|----------------|---------------|------------------------|----------------------|---------------|---------------|-----------|--------------|
| 1              | M             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 2              | E             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 3              | E             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 4              | P             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 5              | Q             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 6              | S             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 7              | D             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 8              | P             | transactivation domain | FALSE                | 56            | 54            | 2         | 0.035714286  |
| 9              | S             | transactivation domain | FALSE                | 56            | 50            | 6         | 0.107142857  |
| 10             | V             | transactivation domain | FALSE                | 56            | 51            | 5         | 0.089285714  |
| 11             | E             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 12             | P             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 13             | P             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 14             | L             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 15             | S             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 16             | Q             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 17             | E             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 18             | T             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 19             | F             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 20             | S             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 21             | D             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 22             | L             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 23             | W             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 24             | K             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 25             | L             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 26             | L             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 27             | P             | transactivation domain | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 28             | E             | transactivation domain | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 29             | N             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 30             | N             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 31             | V             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 32             | L             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 33             | S             | transactivation domain | FALSE                | 56            | 54            | 2         | 0.035714286  |
| 34             | P             | transactivation domain | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 35             | L             | transactivation domain | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 36             | P             | transactivation domain | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 37             | S             | transactivation domain | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 38             | Q             | transactivation domain | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 39             | A             | transactivation domain | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 40             | M             | transactivation domain | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 41             | D             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 42             | D             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 43             | L             | transactivation domain | FALSE                | 56            | 54            | 2         | 0.035714286  |
| 44             | M             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 45             | L             | transactivation domain | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 46             | S             | transactivation domain | FALSE                | 56            | 54            | 2         | 0.035714286  |
| 47             | P             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 48             | D             | transactivation domain | FALSE                | 56            | 22            | 34        | 0.607142857  |
| 49             | D             | transactivation domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 50             | I             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 51             | E             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 52             | Q             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 53             | W             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 54             | F             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 55             | T             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 56             | E             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 57             | D             | transactivation domain | FALSE                | 56            | 44            | 12        | 0.214285714  |
| 58             | P             | transactivation domain | FALSE                | 56            | 7             | 49        | 0.875        |
| 59             | G             | transactivation domain | FALSE                | 56            | 7             | 49        | 0.875        |
| 60             | P             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 61             | D             | transactivation domain | FALSE                | 56            | 56            | 0         | 0            |
| 62             | E             | proline rich region    | FALSE                | 56            | 56            | 0         | 0            |
| 63             | A             | proline rich region    | FALSE                | 56            | 51            | 5         | 0.089285714  |
| 64             | P             | proline rich region    | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 65             | R             | proline rich region    | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 66             | M             | proline rich region    | FALSE                | 56            | 52            | 4         | 0.071428571  |
| 67             | P             | proline rich region    | FALSE                | 56            | 41            | 15        | 0.267857143  |
| 68             | E             | proline rich region    | FALSE                | 56            | 40            | 16        | 0.285714286  |
| 69             | A             | proline rich region    | FALSE                | 56            | 41            | 15        | 0.267857143  |
| 70             | A             | proline rich region    | FALSE                | 56            | 37            | 19        | 0.339285714  |
| 71             | P             | proline rich region    | FALSE                | 56            | 19            | 37        | 0.660714286  |
| 72             | P             | proline rich region    | FALSE                | 56            | 18            | 38        | 0.678571429  |
| 73             | V             | proline rich region    | FALSE                | 56            | 18            | 38        | 0.678571429  |



| human position | human residue | domain                     | is canonical hotspot | total species | non gap count | gap count | gap fraction |
|----------------|---------------|----------------------------|----------------------|---------------|---------------|-----------|--------------|
| 74             | A             | proline rich region        | FALSE                | 56            | 18            | 38        | 0.678571429  |
| 75             | P             | proline rich region        | FALSE                | 56            | 20            | 36        | 0.642857143  |
| 76             | A             | proline rich region        | FALSE                | 56            | 36            | 20        | 0.357142857  |
| 77             | P             | proline rich region        | FALSE                | 56            | 45            | 11        | 0.196428571  |
| 78             | A             | proline rich region        | FALSE                | 56            | 45            | 11        | 0.196428571  |
| 79             | A             | proline rich region        | FALSE                | 56            | 51            | 5         | 0.089285714  |
| 80             | P             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 81             | T             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 82             | P             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 83             | A             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 84             | A             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 85             | P             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 86             | A             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 87             | P             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 88             | A             | proline rich region        | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 89             | P             | proline rich region        | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 90             | S             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 91             | W             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 92             | P             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 93             | L             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 94             | S             | proline rich region        | FALSE                | 56            | 56            | 0         | 0            |
| 95             | S             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 96             | S             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 97             | V             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 98             | P             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 99             | S             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 100            | Q             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 101            | K             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 102            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 103            | Y             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 104            | Q             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 105            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 106            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 107            | Y             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 108            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 109            | F             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 110            | R             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 111            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 112            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 113            | F             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 114            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 115            | H             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 116            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 117            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 118            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 119            | A             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 120            | K             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 121            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 122            | V             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 123            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 124            | C             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 125            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 126            | Y             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 127            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 128            | P             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 129            | A             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 130            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 131            | N             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 132            | K             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 133            | M             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 134            | F             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 135            | C             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 136            | Q             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 137            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 138            | A             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 139            | K             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 140            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 141            | C             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 142            | P             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 143            | V             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 144            | Q             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 145            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 146            | W             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 147            | V             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 148            | D             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |

| human position | human residue | domain             | is canonical hotspot | total species | non gap count | gap count | gap fraction |
|----------------|---------------|--------------------|----------------------|---------------|---------------|-----------|--------------|
| 149            | S             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 150            | T             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 151            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 152            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 153            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 154            | G             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 155            | T             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 156            | R             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 157            | V             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 158            | R             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 159            | A             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 160            | M             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 161            | A             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 162            | I             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 163            | Y             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 164            | K             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 165            | Q             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 166            | S             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 167            | Q             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 168            | H             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 169            | M             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 170            | T             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 171            | E             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 172            | V             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 173            | V             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 174            | R             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 175            | R             | DNA binding domain | TRUE                 | 56            | 56            | 0         | 0            |
| 176            | C             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 177            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 178            | H             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 179            | H             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 180            | E             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 181            | R             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 182            | C             | DNA binding domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 183            | S             | DNA binding domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 184            | D             | DNA binding domain | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 185            | S             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 186            | D             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 187            | G             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 188            | L             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 189            | A             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 190            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 191            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 192            | Q             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 193            | H             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 194            | L             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 195            | I             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 196            | R             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 197            | V             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 198            | E             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 199            | G             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 200            | N             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 201            | L             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 202            | R             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 203            | V             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 204            | E             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 205            | Y             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 206            | L             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 207            | D             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 208            | D             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 209            | R             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 210            | N             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 211            | T             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 212            | F             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 213            | R             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 214            | H             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 215            | S             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 216            | V             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 217            | V             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 218            | V             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 219            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 220            | Y             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 221            | E             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 222            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |
| 223            | P             | DNA binding domain | FALSE                | 56            | 56            | 0         | 0            |

| human position | human residue | domain                     | is canonical hotspot | total species | non gap count | gap count | gap fraction |
|----------------|---------------|----------------------------|----------------------|---------------|---------------|-----------|--------------|
| 224            | E             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 225            | V             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 226            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 227            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 228            | D             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 229            | C             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 230            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 231            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 232            | I             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 233            | H             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 234            | Y             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 235            | N             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 236            | Y             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 237            | M             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 238            | C             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 239            | N             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 240            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 241            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 242            | C             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 243            | M             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 244            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 245            | G             | DNA binding domain         | TRUE                 | 56            | 56            | 0         | 0            |
| 246            | M             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 247            | N             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 248            | R             | DNA binding domain         | TRUE                 | 56            | 56            | 0         | 0            |
| 249            | R             | DNA binding domain         | TRUE                 | 56            | 56            | 0         | 0            |
| 250            | P             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 251            | I             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 252            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 253            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 254            | I             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 255            | I             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 256            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 257            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 258            | E             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 259            | D             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 260            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 261            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 262            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 263            | N             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 264            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 265            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 266            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 267            | R             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 268            | N             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 269            | S             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 270            | F             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 271            | E             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 272            | V             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 273            | R             | DNA binding domain         | TRUE                 | 56            | 56            | 0         | 0            |
| 274            | V             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 275            | C             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 276            | A             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 277            | C             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 278            | P             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 279            | G             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 280            | R             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 281            | D             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 282            | R             | DNA binding domain         | TRUE                 | 56            | 56            | 0         | 0            |
| 283            | R             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 284            | T             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 285            | E             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 286            | E             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 287            | E             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 288            | N             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 289            | L             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 290            | R             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 291            | K             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 292            | K             | DNA binding domain         | FALSE                | 56            | 56            | 0         | 0            |
| 293            | G             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 294            | E             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 295            | P             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 296            | H             | interdomain or unannotated | FALSE                | 56            | 56            | 0         | 0            |
| 297            | H             | interdomain or unannotated | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 298            | E             | interdomain or unannotated | FALSE                | 56            | 55            | 1         | 0.017857143  |

| human position | human residue | domain                       | is canonical hotspot | total species | non gap count | gap count | gap fraction |
|----------------|---------------|------------------------------|----------------------|---------------|---------------|-----------|--------------|
| 299            | L             | interdomain or unannotated   | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 300            | P             | interdomain or unannotated   | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 301            | P             | interdomain or unannotated   | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 302            | G             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 303            | S             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 304            | T             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 305            | K             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 306            | R             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 307            | A             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 308            | L             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 309            | P             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 310            | N             | interdomain or unannotated   | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 311            | N             | interdomain or unannotated   | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 312            | T             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 313            | S             | interdomain or unannotated   | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 314            | S             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 315            | S             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 316            | P             | interdomain or unannotated   | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 317            | Q             | interdomain or unannotated   | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 318            | P             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 319            | K             | interdomain or unannotated   | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 320            | K             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 321            | K             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 322            | P             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 323            | L             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 324            | D             | interdomain or unannotated   | FALSE                | 56            | 56            | 0         | 0            |
| 325            | G             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 326            | E             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 327            | Y             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 328            | F             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 329            | T             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 330            | L             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 331            | Q             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 332            | I             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 333            | R             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 334            | G             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 335            | R             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 336            | E             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 337            | R             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 338            | F             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 339            | E             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 340            | M             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 341            | F             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 342            | R             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 343            | E             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 344            | L             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 345            | N             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 346            | E             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 347            | A             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 348            | L             | oligomerization domain       | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 349            | E             | oligomerization domain       | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 350            | L             | oligomerization domain       | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 351            | K             | oligomerization domain       | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 352            | D             | oligomerization domain       | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 353            | A             | oligomerization domain       | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 354            | Q             | oligomerization domain       | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 355            | A             | oligomerization domain       | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 356            | G             | oligomerization domain       | FALSE                | 56            | 56            | 0         | 0            |
| 357            | K             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 358            | E             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 359            | P             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 360            | G             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 361            | G             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 362            | S             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 363            | R             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 364            | A             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 365            | H             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 366            | S             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 367            | S             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 368            | H             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 369            | L             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 370            | K             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 371            | S             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 372            | K             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 373            | K             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |

| human position | human residue | domain                       | is canonical hotspot | total species | non gap count | gap count | gap fraction |
|----------------|---------------|------------------------------|----------------------|---------------|---------------|-----------|--------------|
| 374            | G             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 375            | Q             | C terminal regulatory region | FALSE                | 56            | 55            | 1         | 0.017857143  |
| 376            | S             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 377            | T             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 378            | S             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 379            | R             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 380            | H             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 381            | K             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 382            | K             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |
| 383            | L             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 384            | M             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 385            | F             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 386            | K             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 387            | T             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 388            | E             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 389            | G             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 390            | P             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 391            | D             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 392            | S             | C terminal regulatory region | FALSE                | 56            | 56            | 0         | 0            |
| 393            | D             | C terminal regulatory region | FALSE                | 56            | 53            | 3         | 0.053571429  |

**Table S3.** Residue-level conservation metrics, part 2 of 2: residue conservation, entropy and amino-acid diversity fields.

| human position | majority residue | majority count | majority conservation | human residue conservation | shannon entropy | normalized entropy | unique residue count |
|----------------|------------------|----------------|-----------------------|----------------------------|-----------------|--------------------|----------------------|
| 1              | M                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 2              | E                | 43             | 0.767857143           | 0.767857143                | 1.004872334     | 0.232505565        | 3                    |
| 3              | E                | 44             | 0.785714286           | 0.785714286                | 1.053515793     | 0.243760602        | 5                    |
| 4              | P                | 30             | 0.535714286           | 0.535714286                | 1.581065134     | 0.365824026        | 5                    |
| 5              | Q                | 44             | 0.785714286           | 0.785714286                | 1.024024999     | 0.236937075        | 4                    |
| 6              | S                | 40             | 0.727272727           | 0.727272727                | 1.269335417     | 0.293696561        | 5                    |
| 7              | E                | 30             | 0.545454545           | 0.454545455                | 0.994030211     | 0.229996934        | 2                    |
| 8              | L                | 44             | 0.814814815           | 0.111111111                | 0.982206646     | 0.227261219        | 5                    |
| 9              | S                | 23             | 0.46                  | 0.46                       | 2.08223569      | 0.481783973        | 7                    |
| 10             | I                | 24             | 0.470588235           | 0.352941176                | 1.753746174     | 0.405778656        | 5                    |
| 11             | E                | 44             | 0.8                   | 0.8                        | 0.809827492     | 0.187376438        | 3                    |
| 12             | P                | 43             | 0.781818182           | 0.781818182                | 1.016191133     | 0.235124488        | 4                    |
| 13             | P                | 54             | 0.981818182           | 0.981818182                | 0.13110653      | 0.030335195        | 2                    |
| 14             | L                | 55             | 1                     | 1                          | 0               | 0                  | 1                    |
| 15             | S                | 55             | 1                     | 1                          | 0               | 0                  | 1                    |
| 16             | Q                | 54             | 0.981818182           | 0.981818182                | 0.13110653      | 0.030335195        | 2                    |
| 17             | E                | 54             | 0.981818182           | 0.981818182                | 0.13110653      | 0.030335195        | 2                    |
| 18             | T                | 54             | 0.964285714           | 0.964285714                | 0.222284831     | 0.051431867        | 2                    |
| 19             | F                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 20             | S                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 21             | D                | 40             | 0.714285714           | 0.714285714                | 1.241537795     | 0.287264797        | 5                    |
| 22             | L                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 23             | W                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 24             | K                | 31             | 0.553571429           | 0.553571429                | 1.400684654     | 0.324087912        | 4                    |
| 25             | L                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 26             | L                | 52             | 0.928571429           | 0.928571429                | 0.371232327     | 0.085895072        | 2                    |
| 27             | P                | 50             | 0.961538462           | 0.961538462                | 0.235193382     | 0.054418624        | 2                    |
| 28             | E                | 38             | 0.730769231           | 0.730769231                | 1.396867151     | 0.323204625        | 6                    |
| 29             | N                | 45             | 0.803571429           | 0.803571429                | 1.144258684     | 0.26475653         | 6                    |
| 30             | N                | 46             | 0.821428571           | 0.821428571                | 0.977919601     | 0.22626929         | 5                    |
| 31             | V                | 30             | 0.545454545           | 0.545454545                | 1.709330525     | 0.395501843        | 6                    |
| 32             | L                | 53             | 0.963636364           | 0.963636364                | 0.261727275     | 0.060557989        | 3                    |
| 33             | S                | 40             | 0.740740741           | 0.740740741                | 1.257162987     | 0.290880126        | 5                    |
| 34             | S                | 20             | 0.377358491           | 0.358490566                | 1.984329708     | 0.459130662        | 6                    |
| 35             | E                | 23             | 0.442307692           | 0.134615385                | 2.13299843      | 0.493529365        | 6                    |
| 36             | L                | 37             | 0.711538462           | 0.134615385                | 1.500415507     | 0.347163459        | 7                    |
| 37             | S                | 47             | 0.886792453           | 0.886792453                | 0.674702275     | 0.156111407        | 4                    |
| 38             | P                | 24             | 0.461538462           | 0.173076923                | 2.374806006     | 0.54947837         | 8                    |
| 39             | A                | 30             | 0.566037736           | 0.566037736                | 1.816365092     | 0.420267309        | 7                    |
| 40             | V                | 32             | 0.615384615           | 0.25                       | 1.544560558     | 0.357377662        | 6                    |
| 41             | D                | 40             | 0.727272727           | 0.727272727                | 1.371748654     | 0.317392753        | 6                    |
| 42             | D                | 29             | 0.527272727           | 0.527272727                | 1.833665083     | 0.424270151        | 8                    |
| 43             | L                | 46             | 0.851851852           | 0.851851852                | 0.864445836     | 0.200013933        | 5                    |
| 44             | L                | 36             | 0.654545455           | 0.127272727                | 1.531018177     | 0.35424425         | 5                    |
| 45             | L                | 41             | 0.773584906           | 0.773584906                | 1.26726914      | 0.293218469        | 6                    |
| 46             | S                | 38             | 0.703703704           | 0.703703704                | 1.49236652      | 0.345301099        | 7                    |
| 47             | P                | 28             | 0.509090909           | 0.509090909                | 2.051627662     | 0.474701943        | 7                    |
| 48             | E                | 13             | 0.590909091           | 0.318181818                | 1.379558855     | 0.319199863        | 4                    |
| 49             | D                | 40             | 0.727272727           | 0.727272727                | 1.469690872     | 0.340054448        | 7                    |
| 50             | V                | 33             | 0.589285714           | 0.089285714                | 2.133674014     | 0.493685681        | 10                   |
| 51             | A                | 22             | 0.392857143           | 0.142857143                | 2.288391548     | 0.529483947        | 8                    |
| 52             | N                | 32             | 0.571428571           | 0.125                      | 2.056321772     | 0.475788057        | 8                    |
| 53             | W                | 51             | 0.910714286           | 0.910714286                | 0.569967557     | 0.131878075        | 4                    |
| 54             | L                | 46             | 0.821428571           | 0.142857143                | 0.805857601     | 0.186457892        | 3                    |
| 55             | D                | 29             | 0.517857143           | 0.107142857                | 2.039133237     | 0.471811005        | 8                    |
| 56             | E                | 35             | 0.625                 | 0.625                      | 1.902992046     | 0.440310899        | 8                    |
| 57             | D                | 13             | 0.295454545           | 0.295454545                | 2.748086943     | 0.635847446        | 9                    |
| 58             | P                | 7              | 1                     | 1                          | 0               | 0                  | 1                    |
| 59             | G                | 6              | 0.857142857           | 0.857142857                | 0.591672779     | 0.13690019         | 2                    |
| 60             | P                | 36             | 0.642857143           | 0.642857143                | 1.669373841     | 0.386256736        | 6                    |
| 61             | D                | 33             | 0.589285714           | 0.589285714                | 1.805736014     | 0.417807972        | 7                    |
| 62             | E                | 44             | 0.785714286           | 0.785714286                | 0.972634989     | 0.225046546        | 4                    |
| 63             | A                | 39             | 0.764705882           | 0.764705882                | 1.419767028     | 0.328503158        | 8                    |
| 64             | P                | 30             | 0.576923077           | 0.576923077                | 2.001055539     | 0.463000655        | 8                    |
| 65             | R                | 25             | 0.480769231           | 0.480769231                | 2.151081082     | 0.497713297        | 8                    |
| 66             | M                | 36             | 0.692307692           | 0.692307692                | 1.298017884     | 0.300333059        | 5                    |
| 67             | P                | 18             | 0.43902439            | 0.43902439                 | 1.734490582     | 0.401323332        | 6                    |
| 68             | E                | 20             | 0.5                   | 0.5                        | 1.639448951     | 0.379332769        | 4                    |
| 69             | A                | 18             | 0.43902439            | 0.43902439                 | 2.199578204     | 0.508934475        | 8                    |
| 70             | P                | 26             | 0.702702703           | 0.216216216                | 1.203738969     | 0.278518972        | 4                    |
| 71             | P                | 7              | 0.368421053           | 0.368421053                | 2.247084754     | 0.519926455        | 6                    |
| 72             | P                | 12             | 0.666666667           | 0.666666667                | 1.134969975     | 0.262607325        | 3                    |

| human position | majority residue | majority count | majority conservation | human residue conservation | shannon entropy | normalized entropy | unique residue count |
|----------------|------------------|----------------|-----------------------|----------------------------|-----------------|--------------------|----------------------|
| 73             | E                | 7              | 0.388888889           | 0.222222222                | 2.258459893     | 0.522558414        | 6                    |
| 74             | A                | 11             | 0.611111111           | 0.611111111                | 1.791667484     | 0.414552821        | 6                    |
| 75             | P                | 18             | 0.9                   | 0.9                        | 0.568995594     | 0.131653184        | 3                    |
| 76             | A                | 24             | 0.666666667           | 0.666666667                | 1.359966238     | 0.314666558        | 4                    |
| 77             | P                | 36             | 0.8                   | 0.8                        | 1.05347664      | 0.243751543        | 5                    |
| 78             | A                | 32             | 0.711111111           | 0.711111111                | 1.456083308     | 0.336905954        | 6                    |
| 79             | A                | 33             | 0.647058824           | 0.647058824                | 1.425940591     | 0.329931586        | 6                    |
| 80             | P                | 51             | 0.910714286           | 0.910714286                | 0.556487423     | 0.128759066        | 4                    |
| 81             | A                | 29             | 0.517857143           | 0.267857143                | 2.039594515     | 0.471917734        | 9                    |
| 82             | P                | 42             | 0.75                  | 0.75                       | 1.307585158     | 0.302546718        | 6                    |
| 83             | A                | 34             | 0.607142857           | 0.607142857                | 1.628903375     | 0.376892752        | 5                    |
| 84             | A                | 41             | 0.732142857           | 0.732142857                | 1.391084982     | 0.321866757        | 7                    |
| 85             | P                | 47             | 0.839285714           | 0.839285714                | 0.783598877     | 0.181307708        | 3                    |
| 86             | A                | 41             | 0.732142857           | 0.732142857                | 1.364871928     | 0.315801628        | 6                    |
| 87             | P                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 88             | A                | 48             | 0.872727273           | 0.872727273                | 0.72539244      | 0.167840007        | 4                    |
| 89             | T                | 33             | 0.6                   | 0.2                        | 1.706298        | 0.394800182        | 6                    |
| 90             | S                | 44             | 0.785714286           | 0.785714286                | 1.081973437     | 0.25034508         | 5                    |
| 91             | W                | 52             | 0.928571429           | 0.928571429                | 0.371232327     | 0.085895072        | 2                    |
| 92             | P                | 51             | 0.910714286           | 0.910714286                | 0.43408112      | 0.100436914        | 2                    |
| 93             | L                | 52             | 0.928571429           | 0.928571429                | 0.371232327     | 0.085895072        | 2                    |
| 94             | S                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 95             | S                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 96             | F                | 26             | 0.464285714           | 0.446428571                | 1.344542817     | 0.311097914        | 3                    |
| 97             | V                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 98             | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 99             | S                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 100            | Q                | 44             | 0.785714286           | 0.785714286                | 1.095453571     | 0.25346409         | 5                    |
| 101            | K                | 52             | 0.928571429           | 0.928571429                | 0.429180764     | 0.099303078        | 3                    |
| 102            | T                | 48             | 0.857142857           | 0.857142857                | 0.877387064     | 0.203008251        | 6                    |
| 103            | Y                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 104            | P                | 40             | 0.714285714           | 0.160714286                | 1.289210471     | 0.298295215        | 5                    |
| 105            | G                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 106            | S                | 20             | 0.357142857           | 0.357142857                | 2.520281375     | 0.583138201        | 8                    |
| 107            | Y                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 108            | G                | 51             | 0.910714286           | 0.910714286                | 0.556487423     | 0.128759066        | 4                    |
| 109            | F                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 110            | R                | 47             | 0.839285714           | 0.839285714                | 0.794548485     | 0.183841209        | 4                    |
| 111            | L                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 112            | G                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 113            | F                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 114            | L                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 115            | H                | 33             | 0.589285714           | 0.589285714                | 1.58910123      | 0.367683403        | 5                    |
| 116            | S                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 117            | G                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 118            | T                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 119            | A                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 120            | K                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 121            | S                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 122            | V                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 123            | T                | 54             | 0.964285714           | 0.964285714                | 0.222284831     | 0.051431867        | 2                    |
| 124            | C                | 54             | 0.964285714           | 0.964285714                | 0.222284831     | 0.051431867        | 2                    |
| 125            | T                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 126            | Y                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 127            | S                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 128            | P                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 129            | S                | 12             | 0.214285714           | 0.196428571                | 3.043714167     | 0.704249145        | 12                   |
| 130            | L                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 131            | N                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 132            | K                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 133            | L                | 48             | 0.857142857           | 0.142857143                | 0.591672779     | 0.13690019         | 2                    |
| 134            | F                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 135            | C                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 136            | Q                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 137            | L                | 54             | 0.964285714           | 0.964285714                | 0.222284831     | 0.051431867        | 2                    |
| 138            | A                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 139            | K                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 140            | T                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 141            | C                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 142            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 143            | V                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 144            | Q                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 145            | L                | 54             | 0.964285714           | 0.964285714                | 0.222284831     | 0.051431867        | 2                    |
| 146            | W                | 52             | 0.928571429           | 0.928571429                | 0.429180764     | 0.099303078        | 3                    |



| human position | majority residue | majority count | majority conservation | human residue conservation | shannon entropy | normalized entropy | unique residue count |
|----------------|------------------|----------------|-----------------------|----------------------------|-----------------|--------------------|----------------------|
| 147            | V                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 148            | S                | 21             | 0.375                 | 0.267857143                | 2.304765748     | 0.533272581        | 7                    |
| 149            | S                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 150            | P                | 38             | 0.678571429           | 0.285714286                | 1.067689824     | 0.247040164        | 3                    |
| 151            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 152            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 153            | P                | 41             | 0.732142857           | 0.732142857                | 1.465044079     | 0.338979281        | 8                    |
| 154            | G                | 48             | 0.857142857           | 0.857142857                | 0.792478359     | 0.183362227        | 4                    |
| 155            | T                | 42             | 0.75                  | 0.75                       | 1.057585158     | 0.244702164        | 3                    |
| 156            | R                | 47             | 0.839285714           | 0.839285714                | 0.716896422     | 0.165874213        | 3                    |
| 157            | V                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 158            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 159            | A                | 53             | 0.946428571           | 0.946428571                | 0.301378644     | 0.069732452        | 2                    |
| 160            | M                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 161            | A                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 162            | I                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 163            | Y                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 164            | K                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 165            | K                | 48             | 0.857142857           | 0.142857143                | 0.591672779     | 0.13690019         | 2                    |
| 166            | S                | 52             | 0.928571429           | 0.928571429                | 0.371232327     | 0.085895072        | 2                    |
| 167            | E                | 38             | 0.678571429           | 0.321428571                | 0.905928216     | 0.209612052        | 2                    |
| 168            | H                | 32             | 0.571428571           | 0.571428571                | 1.411650205     | 0.326625102        | 3                    |
| 169            | M                | 49             | 0.875                 | 0.875                      | 0.543564443     | 0.12576897         | 2                    |
| 170            | T                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 171            | E                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 172            | V                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 173            | V                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 174            | R                | 48             | 0.857142857           | 0.857142857                | 0.591672779     | 0.13690019         | 2                    |
| 175            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 176            | C                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 177            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 178            | H                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 179            | H                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 180            | E                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 181            | R                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 182            | C                | 48             | 0.872727273           | 0.872727273                | 0.549910905     | 0.127237403        | 2                    |
| 183            | S                | 47             | 0.854545455           | 0.854545455                | 0.78726305      | 0.182155518        | 4                    |
| 184            | D                | 49             | 0.890909091           | 0.890909091                | 0.568079298     | 0.131441173        | 3                    |
| 185            | S                | 43             | 0.767857143           | 0.767857143                | 1.219158048     | 0.282086611        | 5                    |
| 186            | D                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 187            | G                | 51             | 0.910714286           | 0.910714286                | 0.556487423     | 0.128759066        | 4                    |
| 188            | L                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 189            | A                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 190            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 191            | P                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 192            | Q                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 193            | H                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 194            | L                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 195            | I                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 196            | R                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 197            | V                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 198            | E                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 199            | G                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 200            | N                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 201            | L                | 53             | 0.946428571           | 0.946428571                | 0.386287349     | 0.089378477        | 4                    |
| 202            | R                | 43             | 0.767857143           | 0.767857143                | 1.01184304      | 0.234118435        | 4                    |
| 203            | A                | 47             | 0.839285714           | 0.142857143                | 0.716896422     | 0.165874213        | 3                    |
| 204            | E                | 43             | 0.767857143           | 0.767857143                | 0.925513275     | 0.214143608        | 3                    |
| 205            | Y                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 206            | L                | 50             | 0.892857143           | 0.892857143                | 0.661054753     | 0.152953667        | 4                    |
| 207            | D                | 44             | 0.785714286           | 0.785714286                | 1.120218248     | 0.259194097        | 5                    |
| 208            | D                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 209            | R                | 37             | 0.660714286           | 0.660714286                | 1.847188412     | 0.427399154        | 9                    |
| 210            | N                | 38             | 0.678571429           | 0.678571429                | 1.647872413     | 0.381281774        | 7                    |
| 211            | T                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 212            | F                | 47             | 0.839285714           | 0.839285714                | 0.931182136     | 0.215455259        | 5                    |
| 213            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 214            | H                | 51             | 0.910714286           | 0.910714286                | 0.43408112      | 0.100436914        | 2                    |
| 215            | S                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 216            | V                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 217            | V                | 52             | 0.928571429           | 0.928571429                | 0.371232327     | 0.085895072        | 2                    |
| 218            | V                | 54             | 0.964285714           | 0.964285714                | 0.222284831     | 0.051431867        | 2                    |
| 219            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 220            | Y                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |



| human position | majority residue | majority count | majority conservation | human residue conservation | shannon entropy | normalized entropy | unique residue count |
|----------------|------------------|----------------|-----------------------|----------------------------|-----------------|--------------------|----------------------|
| 221            | E                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 222            | P                | 45             | 0.803571429           | 0.803571429                | 1.032389711     | 0.238872487        | 5                    |
| 223            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 224            | E                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 225            | V                | 51             | 0.910714286           | 0.910714286                | 0.556487423     | 0.128759066        | 4                    |
| 226            | G                | 52             | 0.928571429           | 0.928571429                | 0.442660898     | 0.102422088        | 3                    |
| 227            | S                | 51             | 0.910714286           | 0.910714286                | 0.498538986     | 0.11535106         | 3                    |
| 228            | D                | 49             | 0.875                 | 0.875                      | 0.543564443     | 0.12576897         | 2                    |
| 229            | C                | 48             | 0.857142857           | 0.857142857                | 0.591672779     | 0.13690019         | 2                    |
| 230            | T                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 231            | T                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 232            | I                | 51             | 0.910714286           | 0.910714286                | 0.43408112      | 0.100436914        | 2                    |
| 233            | H                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 234            | Y                | 53             | 0.946428571           | 0.946428571                | 0.301378644     | 0.069732452        | 2                    |
| 235            | N                | 54             | 0.964285714           | 0.964285714                | 0.222284831     | 0.051431867        | 2                    |
| 236            | Y                | 29             | 0.517857143           | 0.517857143                | 0.999079718     | 0.23116528         | 2                    |
| 237            | M                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 238            | C                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 239            | N                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 240            | S                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 241            | S                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 242            | C                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 243            | M                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 244            | G                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 245            | G                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 246            | M                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 247            | N                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 248            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 249            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 250            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 251            | I                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 252            | L                | 50             | 0.892857143           | 0.892857143                | 0.589626181     | 0.136426652        | 3                    |
| 253            | T                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 254            | I                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 255            | I                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 256            | T                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 257            | L                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 258            | E                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 259            | D                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 260            | S                | 51             | 0.910714286           | 0.910714286                | 0.520773137     | 0.120495558        | 3                    |
| 261            | S                | 40             | 0.714285714           | 0.714285714                | 1.327867561     | 0.307239623        | 6                    |
| 262            | G                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 263            | N                | 45             | 0.803571429           | 0.803571429                | 1.072830113     | 0.248229514        | 5                    |
| 264            | L                | 48             | 0.857142857           | 0.857142857                | 0.591672779     | 0.13690019         | 2                    |
| 265            | L                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 266            | G                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 267            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 268            | N                | 39             | 0.696428571           | 0.696428571                | 1.43485952      | 0.331995232        | 5                    |
| 269            | S                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 270            | F                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 271            | E                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 272            | V                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 273            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 274            | V                | 49             | 0.875                 | 0.875                      | 0.543564443     | 0.12576897         | 2                    |
| 275            | C                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 276            | A                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 277            | C                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 278            | P                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 279            | G                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 280            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 281            | D                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 282            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 283            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 284            | T                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 285            | E                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 286            | E                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 287            | E                | 52             | 0.928571429           | 0.928571429                | 0.429180764     | 0.099303078        | 3                    |
| 288            | N                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 289            | F                | 51             | 0.910714286           | 0.089285714                | 0.43408112      | 0.100436914        | 2                    |
| 290            | R                | 41             | 0.732142857           | 0.732142857                | 1.080940954     | 0.250106187        | 4                    |
| 291            | K                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 292            | K                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 293            | G                | 49             | 0.875                 | 0.875                      | 0.617523541     | 0.142881493        | 3                    |
| 294            | E                | 36             | 0.642857143           | 0.642857143                | 1.438665189     | 0.332875781        | 5                    |

| human position | majority residue | majority count | majority conservation | human residue conservation | shannon entropy | normalized entropy | unique residue count |
|----------------|------------------|----------------|-----------------------|----------------------------|-----------------|--------------------|----------------------|
| 295            | P                | 34             | 0.607142857           | 0.607142857                | 1.493478216     | 0.345558321        | 5                    |
| 296            | C                | 41             | 0.732142857           | 0.071428571                | 1.414066348     | 0.327184145        | 6                    |
| 297            | P                | 47             | 0.854545455           | 0.109090909                | 0.752715149     | 0.174161886        | 4                    |
| 298            | E                | 41             | 0.745454545           | 0.745454545                | 1.187192764     | 0.274690541        | 4                    |
| 299            | P                | 31             | 0.563636364           | 0.327272727                | 1.478837247     | 0.34217072         | 5                    |
| 300            | P                | 50             | 0.909090909           | 0.909090909                | 0.527765223     | 0.122113374        | 3                    |
| 301            | P                | 44             | 0.8                   | 0.8                        | 1.009002373     | 0.233461166        | 4                    |
| 302            | G                | 47             | 0.839285714           | 0.839285714                | 0.859753564     | 0.198928243        | 4                    |
| 303            | S                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 304            | T                | 37             | 0.660714286           | 0.660714286                | 1.508713083     | 0.349083337        | 5                    |
| 305            | K                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 306            | R                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 307            | A                | 50             | 0.892857143           | 0.892857143                | 0.560882601     | 0.129776014        | 3                    |
| 308            | L                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 309            | P                | 52             | 0.928571429           | 0.928571429                | 0.429180764     | 0.099303078        | 3                    |
| 310            | T                | 34             | 0.618181818           | 0.127272727                | 1.614913421     | 0.373655782        | 5                    |
| 311            | S                | 21             | 0.381818182           | 0.345454545                | 2.036471738     | 0.471195192        | 6                    |
| 312            | T                | 50             | 0.892857143           | 0.892857143                | 0.589626181     | 0.136426652        | 3                    |
| 313            | S                | 45             | 0.818181818           | 0.818181818                | 1.033479907     | 0.239124734        | 5                    |
| 314            | S                | 52             | 0.928571429           | 0.928571429                | 0.478375184     | 0.110685595        | 4                    |
| 315            | S                | 49             | 0.875                 | 0.875                      | 0.617523541     | 0.142881493        | 3                    |
| 316            | P                | 49             | 0.890909091           | 0.890909091                | 0.597345489     | 0.138212732        | 3                    |
| 317            | P                | 24             | 0.452830189           | 0.433962264                | 1.537712436     | 0.355793156        | 5                    |
| 318            | Q                | 25             | 0.446428571           | 0.410714286                | 1.719976043     | 0.397964983        | 6                    |
| 319            | K                | 50             | 0.909090909           | 0.909090909                | 0.439496987     | 0.101690028        | 2                    |
| 320            | K                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 321            | K                | 53             | 0.946428571           | 0.946428571                | 0.301378644     | 0.069732452        | 2                    |
| 322            | P                | 51             | 0.910714286           | 0.910714286                | 0.498538986     | 0.11535106         | 3                    |
| 323            | L                | 47             | 0.839285714           | 0.839285714                | 0.902438556     | 0.208804621        | 5                    |
| 324            | D                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 325            | G                | 47             | 0.839285714           | 0.839285714                | 0.716896422     | 0.165874213        | 3                    |
| 326            | E                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 327            | Y                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 328            | F                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 329            | T                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 330            | L                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 331            | Q                | 46             | 0.821428571           | 0.821428571                | 0.67694187      | 0.1566296          | 2                    |
| 332            | I                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 333            | R                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 334            | G                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 335            | R                | 52             | 0.928571429           | 0.928571429                | 0.514089469     | 0.118949103        | 5                    |
| 336            | E                | 41             | 0.732142857           | 0.732142857                | 1.364124714     | 0.315628739        | 6                    |
| 337            | R                | 52             | 0.928571429           | 0.928571429                | 0.429180764     | 0.099303078        | 3                    |
| 338            | F                | 46             | 0.821428571           | 0.821428571                | 0.760691083     | 0.176007344        | 3                    |
| 339            | E                | 46             | 0.821428571           | 0.821428571                | 0.919971164     | 0.212861284        | 4                    |
| 340            | M                | 48             | 0.857142857           | 0.857142857                | 0.899621216     | 0.208152749        | 6                    |
| 341            | F                | 49             | 0.875                 | 0.875                      | 0.617523541     | 0.142881493        | 3                    |
| 342            | R                | 49             | 0.875                 | 0.875                      | 0.773860817     | 0.179054533        | 5                    |
| 343            | E                | 49             | 0.875                 | 0.875                      | 0.858769523     | 0.198700558        | 7                    |
| 344            | L                | 52             | 0.928571429           | 0.928571429                | 0.371232327     | 0.085895072        | 2                    |
| 345            | N                | 54             | 0.964285714           | 0.964285714                | 0.257999116     | 0.059695375        | 3                    |
| 346            | E                | 53             | 0.946428571           | 0.946428571                | 0.350573063     | 0.081114969        | 3                    |
| 347            | A                | 56             | 1                     | 1                          | 0               | 0                  | 1                    |
| 348            | L                | 55             | 1                     | 1                          | 0               | 0                  | 1                    |
| 349            | E                | 55             | 1                     | 1                          | 0               | 0                  | 1                    |
| 350            | L                | 52             | 0.945454545           | 0.945454545                | 0.305400571     | 0.070663038        | 2                    |
| 351            | K                | 52             | 0.945454545           | 0.945454545                | 0.355489435     | 0.08225251         | 3                    |
| 352            | D                | 53             | 0.963636364           | 0.963636364                | 0.225363639     | 0.052144236        | 2                    |
| 353            | A                | 54             | 0.981818182           | 0.981818182                | 0.13110653      | 0.030335195        | 2                    |
| 354            | Q                | 41             | 0.745454545           | 0.745454545                | 1.148252593     | 0.265680633        | 4                    |
| 355            | A                | 33             | 0.6                   | 0.6                        | 1.669173564     | 0.386210397        | 6                    |
| 356            | G                | 44             | 0.785714286           | 0.785714286                | 1.155932534     | 0.267457604        | 6                    |
| 357            | K                | 48             | 0.857142857           | 0.857142857                | 0.777214913     | 0.179830598        | 4                    |
| 358            | E                | 50             | 0.892857143           | 0.892857143                | 0.589626181     | 0.136426652        | 3                    |
| 359            | P                | 48             | 0.872727273           | 0.872727273                | 0.625214713     | 0.144661063        | 3                    |
| 360            | G                | 40             | 0.727272727           | 0.727272727                | 1.234152355     | 0.285555967        | 5                    |
| 361            | G                | 34             | 0.618181818           | 0.618181818                | 1.327291775     | 0.307106399        | 4                    |
| 362            | S                | 50             | 0.892857143           | 0.892857143                | 0.560882601     | 0.129776014        | 3                    |
| 363            | R                | 44             | 0.785714286           | 0.785714286                | 1.081973437     | 0.25034508         | 5                    |
| 364            | A                | 43             | 0.767857143           | 0.767857143                | 1.176473057     | 0.272210234        | 6                    |
| 365            | H                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 366            | S                | 49             | 0.890909091           | 0.890909091                | 0.692711171     | 0.160278273        | 5                    |
| 367            | S                | 53             | 0.963636364           | 0.963636364                | 0.261727275     | 0.060557989        | 3                    |
| 368            | H                | 37             | 0.698113208           | 0.698113208                | 1.336415275     | 0.309217378        | 4                    |

| human position | majority residue | majority count | majority conservation | human residue conservation | shannon entropy | normalized entropy | unique residue count |
|----------------|------------------|----------------|-----------------------|----------------------------|-----------------|--------------------|----------------------|
| 369            | L                | 50             | 0.909090909           | 0.909090909                | 0.577854086     | 0.133702846        | 4                    |
| 370            | K                | 53             | 0.963636364           | 0.963636364                | 0.225363639     | 0.052144236        | 2                    |
| 371            | S                | 42             | 0.763636364           | 0.763636364                | 0.973150619     | 0.225165851        | 3                    |
| 372            | K                | 51             | 0.927272727           | 0.927272727                | 0.48511076      | 0.112244061        | 4                    |
| 373            | K                | 50             | 0.909090909           | 0.909090909                | 0.614217723     | 0.142116599        | 5                    |
| 374            | G                | 51             | 0.927272727           | 0.927272727                | 0.48511076      | 0.112244061        | 4                    |
| 375            | Q                | 45             | 0.818181818           | 0.818181818                | 0.997116271     | 0.230710981        | 5                    |
| 376            | S                | 50             | 0.943396226           | 0.943396226                | 0.403527823     | 0.093367547        | 4                    |
| 377            | T                | 38             | 0.716981132           | 0.716981132                | 1.157545345     | 0.267830774        | 4                    |
| 378            | S                | 48             | 0.905660377           | 0.905660377                | 0.450791388     | 0.104303306        | 2                    |
| 379            | R                | 38             | 0.716981132           | 0.716981132                | 1.323746011     | 0.306285987        | 5                    |
| 380            | H                | 46             | 0.867924528           | 0.867924528                | 0.782942153     | 0.181155756        | 5                    |
| 381            | K                | 52             | 0.981132075           | 0.981132075                | 0.135036203     | 0.031244435        | 2                    |
| 382            | K                | 48             | 0.905660377           | 0.905660377                | 0.518897812     | 0.120061649        | 3                    |
| 383            | L                | 37             | 0.660714286           | 0.660714286                | 1.490045555     | 0.344764078        | 6                    |
| 384            | M                | 41             | 0.732142857           | 0.732142857                | 1.168094249     | 0.27027156         | 4                    |
| 385            | F                | 34             | 0.607142857           | 0.607142857                | 1.530514536     | 0.354127719        | 4                    |
| 386            | K                | 55             | 0.982142857           | 0.982142857                | 0.129233775     | 0.02990188         | 2                    |
| 387            | R                | 38             | 0.678571429           | 0.160714286                | 1.493779726     | 0.345628084        | 6                    |
| 388            | E                | 53             | 0.946428571           | 0.946428571                | 0.350573063     | 0.081114969        | 3                    |
| 389            | G                | 47             | 0.839285714           | 0.839285714                | 0.966896422     | 0.223718766        | 6                    |
| 390            | P                | 49             | 0.875                 | 0.875                      | 0.773860817     | 0.179054533        | 5                    |
| 391            | D                | 53             | 0.946428571           | 0.946428571                | 0.350573063     | 0.081114969        | 3                    |
| 392            | S                | 52             | 0.928571429           | 0.928571429                | 0.478375184     | 0.110685595        | 4                    |
| 393            | D                | 52             | 0.981132075           | 0.981132075                | 0.135036203     | 0.031244435        | 2                    |

**Table S4.** cBioPortal mutation frequency by codon.

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sa<br>mple<br>co<br>un<br>t | protein changes                                                                               | mutation types                                                                | muta<br>tion<br>frequ<br>ency<br>fracti<br>on | muta<br>tion<br>frequ<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|-----------------------------|
| 273           | 268                           | 22                                           | 267                                           | R273C;R273G;R273H;R273L;R273Lfs*72;R273P;R273S                                                | Frame_Shift_Del;Missense_Mutation                                             | 0.063431953                                   | 6.343195266                                  | 1                           |
| 248           | 225                           | 23                                           | 224                                           | R248G;R248Hfs*13;R248L;R248P;R248Q;R248W                                                      | Frame_Shift_Del;Missense_Mutation                                             | 0.053254438                                   | 5.325443787                                  | 2                           |
| 175           | 167                           | 18                                           | 167                                           | R175C;R175G;R175H;R175L                                                                       | Missense_Mutation                                                             | 0.039526627                                   | 3.952662722                                  | 3                           |
| 213           | 93                            | 20                                           | 92                                            | R213*;R213Dfs*34;R213G;R213L;R213P;R213Q                                                      | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                           | 0.022011834                                   | 2.201183432                                  | 4                           |
| 282           | 91                            | 16                                           | 91                                            | R282G;R282Gfs*63;R282Q;R282W;R282_R283del                                                     | Frame_Shift_Del;In_Frame_Del;Missense_Mutation                                | 0.021538462                                   | 2.153846154                                  | 5                           |
| 245           | 91                            | 17                                           | 89                                            | G245C;G245D;G245R;G245S;G245V                                                                 | Missense_Mutation                                                             | 0.021538462                                   | 2.153846154                                  | 6                           |
| 220           | 78                            | 20                                           | 78                                            | Y220*;Y220C;Y220D;Y220H;Y220Mfs*27;Y220S                                                      | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                           | 0.018461538                                   | 1.846153846                                  | 7                           |
| 179           | 69                            | 17                                           | 69                                            | H179D;H179L;H179N;H179P;H179Q;H179R;H179Y                                                     | Missense_Mutation                                                             | 0.016331361                                   | 1.633136095                                  | 8                           |
| 125           | 66                            | 17                                           | 66                                            | T125=;T125M;T125P;X125_splice                                                                 | Missense_Mutation;Splice_Region;Splice_Site                                   | 0.015621302                                   | 1.562130178                                  | 9                           |
| 193           | 64                            | 17                                           | 64                                            | H193D;H193L;H193N;H193P;H193R;H193Y                                                           | Missense_Mutation                                                             | 0.015147929                                   | 1.514792899                                  | 10                          |
| 331           | 59                            | 16                                           | 59                                            | Q331*;Q331H;Q331Sfs*6;X331_splice                                                             | Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation;Splice_Region;Splice_Site | 0.013964497                                   | 1.396449704                                  | 11                          |
| 176           | 55                            | 16                                           | 55                                            | C176*;C176F;C176G;C176R;C176S;C176Sfs*71;C176W;C176Y;C176dup                                  | Frame_Shift_Del;In_Frame_Ins;Missense_Mutation;Nonsense_Mutation              | 0.013017751                                   | 1.301775148                                  | 12                          |
| 196           | 55                            | 16                                           | 55                                            | R196*;R196P                                                                                   | Missense_Mutation;Nonsense_Mutation                                           | 0.013017751                                   | 1.301775148                                  | 13                          |
| 280           | 52                            | 13                                           | 52                                            | R280*;R280Efs*65;R280G;R280I;R280K;R280Kfs*59;R280S;R280T                                     | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                           | 0.012307692                                   | 1.230769231                                  | 14                          |
| 158           | 52                            | 14                                           | 52                                            | R158Afs*12;R158C;R158G;R158H;R158L;R158P;R158S                                                | Frame_Shift_Del;Missense_Mutation                                             | 0.012307692                                   | 1.230769231                                  | 15                          |
| 126           | 52                            | 18                                           | 51                                            | X126_splice;Y126C;Y126D;Y126H;Y126N;Y126S;Y126_M133del                                        | In_Frame_Del;Missense_Mutation;Splice_Site                                    | 0.012307692                                   | 1.230769231                                  | 16                          |
| 187           | 51                            | 17                                           | 51                                            | G187S;X187_splice                                                                             | Missense_Mutation;Splice_Site                                                 | 0.012071006                                   | 1.207100592                                  | 17                          |
| 278           | 50                            | 14                                           | 50                                            | P278A;P278H;P278L;P278Lfs*67;P278Lfs*68;P278R;P278S;P278T                                     | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation                             | 0.01183432                                    | 1.183431953                                  | 18                          |
| 249           | 47                            | 14                                           | 47                                            | R249Ffs*96;R249G;R249Gfs*96;R249M;R249S;R249T;R249W                                           | Frame_Shift_Del;Missense_Mutation                                             | 0.01112426                                    | 1.112426036                                  | 19                          |
| 157           | 44                            | 12                                           | 44                                            | V157Afs*24;V157D;V157F;V157G;V157Hfs*21;V157L;V157Pfs*23;V157Sfs*13;V157_P177del;V157_R158dup | Frame_Shift_Del;Frame_Shift_Ins;In_Frame_Del;In_Frame_Ins;Missense_Mutation   | 0.010414201                                   | 1.041420118                                  | 20                          |
| 266           | 44                            | 16                                           | 44                                            | G266*;G266Dfs*79;G266E;G266R;G266V                                                            | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                           | 0.010414201                                   | 1.041420118                                  | 21                          |
| 342           | 44                            | 16                                           | 44                                            | R342*;R342Efs*2;R342Efs*3;R342P                                                               | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                           | 0.010414201                                   | 1.041420118                                  | 22                          |
| 195           | 44                            | 16                                           | 44                                            | I195F;I195Ffs*52;I195Lfs*53;I195M;I195N;I195Nfs*14;I195S;I195Sfs*52;I195T;I195Yfs*14          | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation                             | 0.0104142                                     | 1.0414201                                    | 23                          |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes                                                                             | mutation types                                                                                | muta<br>tion<br>fre<br>qu<br>ency<br>frac<br>tion | muta<br>tion<br>fre<br>qu<br>ency<br>per<br>cent | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 241           | 44                            | 18                                           | 44                                            | S241A;S241C;S241F;S241P;S241Pfs*6;S241Y;S241dup                                             | Frame_Shift_Del;In_Frame_Ins;Missense_Mutation                                                | 0.010<br>4142<br>01                               | 18<br>1.041<br>4201<br>18                        | 24                          |
| 275           | 40                            | 17                                           | 40                                            | C275*;C275F;C275G;C275Lfs*67;C275Lfs*70;C275R;C275S;C275Vfs*70;C275W;C275Y;C275_R282delinsW | Frame_Shift_Del;In_Frame_Del;Missense_Mutation;Nonsense_Mutation                              | 0.009<br>4674<br>56                               | 0.946<br>7455<br>62                              | 25                          |
| 238           | 40                            | 18                                           | 40                                            | C238*;C238F;C238G;C238Lfs*9;C238R;C238S;C238W;C238Y;C238_M243del;C238_N239delinsY           | Frame_Shift_Del;In_Frame_Del;Missense_Mutation                                                | 0.009<br>4674<br>56                               | 0.946<br>7455<br>62                              | 26                          |
| 163           | 38                            | 11                                           | 38                                            | Y163*;Y163C;Y163D;Y163H;Y163Lfs*18;Y163N                                                    | Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation                                           | 0.008<br>9940<br>83                               | 0.899<br>4082<br>84                              | 27                          |
| 285           | 37                            | 11                                           | 37                                            | E285*;E285K;E285Q;E285Rfs*54;E285V                                                          | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                                           | 0.008<br>7573<br>96                               | 0.875<br>7396<br>45                              | 28                          |
| 132           | 36                            | 14                                           | 36                                            | K132E;K132M;K132N;K132Q;K132R;K132T;K132Vfs*15                                              | Frame_Shift_Del;Missense_Mutation                                                             | 0.008<br>5207<br>1                                | 0.852<br>0710<br>06                              | 29                          |
| 224           | 36                            | 13                                           | 36                                            | E224*;E224D;E224Gfs*4;X224_splice                                                           | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation;Splice_Region;Splice_Site | 0.008<br>5207<br>1                                | 0.852<br>0710<br>06                              | 30                          |
| 151           | 36                            | 16                                           | 35                                            | P151A;P151H;P151L;P151R;P151Rfs*27;P151S;P151T                                              | Frame_Shift_Del;Missense_Mutation                                                             | 0.008<br>5207<br>1                                | 0.852<br>0710<br>06                              | 31                          |
| 173           | 35                            | 13                                           | 35                                            | V173*;V173A;V173Afs*69;V173Efs*7;V173G;V173L;V173M;V173_R175del;V173dup                     | Frame_Shift_Del;In_Frame_Del;In_Frame_Ins;Missense_Mutation                                   | 0.008<br>2840<br>24                               | 0.828<br>4023<br>67                              | 32                          |
| 237           | 35                            | 15                                           | 35                                            | M237Cfs*10;M237Gfs*20;M237I;M237K;M237L;M237V;M237_N239del                                  | Frame_Shift_Del;In_Frame_Del;Missense_Mutation                                                | 0.008<br>2840<br>24                               | 0.828<br>4023<br>67                              | 33                          |
| 242           | 34                            | 14                                           | 34                                            | C242*;C242Afs*5;C242F;C242G;C242R;C242S;C242Y                                               | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                                           | 0.008<br>0473<br>37                               | 0.804<br>7337<br>28                              | 34                          |
| 205           | 34                            | 16                                           | 34                                            | Y205*;Y205C;Y205D;Y205F;Y205H;Y205N;Y205S                                                   | Missense_Mutation;Nonsense_Mutation                                                           | 0.008<br>0473<br>37                               | 0.804<br>7337<br>28                              | 35                          |
| 286           | 34                            | 15                                           | 34                                            | E286*;E286A;E286G;E286K;E286Q;E286V;E286_E287del                                            | In_Frame_Del;Missense_Mutation;Nonsense_Mutation                                              | 0.008<br>0473<br>37                               | 0.804<br>7337<br>28                              | 36                          |
| 306           | 33                            | 13                                           | 33                                            | R306*;R306Afs*31                                                                            | Frame_Shift_Ins;Nonsense_Mutation                                                             | 0.007<br>8106<br>51                               | 0.781<br>0650<br>89                              | 37                          |
| 225           | 32                            | 13                                           | 32                                            | V225Rfs*23;X225_splice                                                                      | Frame_Shift_Ins;Splice_Site                                                                   | 0.007<br>5739<br>64                               | 0.757<br>3964<br>5                               | 38                          |
| 281           | 32                            | 15                                           | 31                                            | D281A;D281Afs*62;D281Afs*64;D281Afs*66;D281E;D281Efs*26;D281Gfs*63;D281H;D281N;D281V;D281Y  | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation                                             | 0.007<br>5739<br>64                               | 0.757<br>3964<br>5                               | 39                          |
| 135           | 31                            | 15                                           | 31                                            | C135*;C135Afs*35;C135F;C135Lfs*14;C135R;C135W;C135Y                                         | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation                           | 0.007<br>3372<br>78                               | 0.733<br>7278<br>11                              | 40                          |
| 141           | 31                            | 15                                           | 31                                            | C141*;C141Afs*29;C141F;C141G;C141R;C141S;C141W;C141Y                                        | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                                           | 0.007<br>3372<br>78                               | 0.733<br>7278<br>11                              | 41                          |
| 337           | 30                            | 15                                           | 30                                            | R337C;R337H;R337L;R337P;R337S                                                               | Missense_Mutation                                                                             | 0.007<br>1005<br>92                               | 0.710<br>0591<br>72                              | 42                          |
| 307           | 30                            | 14                                           | 29                                            | A307Tfs*29;A307V;X307_splice                                                                | Frame_Shift_Del;Missense_Mutation;Splice_Site                                                 | 0.007<br>1005<br>92                               | 0.710<br>0591<br>72                              | 43                          |
| 234           | 29                            | 12                                           | 29                                            | Y234*;Y234C;Y234H;Y234N;Y234Pfs*7;Y234S;Y234Tfs*11                                          | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation                           | 0.006<br>8639<br>05                               | 0.686<br>3905<br>33                              | 44                          |
| 272           | 29                            | 14                                           | 29                                            | V272A;V272Cfs*73;V272G;V272L;V272M                                                          | Frame_Shift_Del;Missense_Mutation                                                             | 0.006<br>8639<br>05                               | 0.686<br>3905<br>33                              | 45                          |
| 244           | 29                            | 13                                           | 29                                            | G244*;G244Afs*3;G244C;G244D;G244R;G244S;G244V;G244_M246delinsV                              | Frame_Shift_Del;In_Frame_Del;Missense_Mutation                                                | 0.006<br>8639<br>05                               | 0.686<br>3905<br>33                              | 46                          |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes                                                                | mutation types                                                                   | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 192           | 27                            | 13                                           | 27                                            | Q192*;Q192R;Q192_H193del;Q192del                                               | In_Frame_Del;Missense_Mutation;Nonsense_Mutation                                 | 0.006390533                                       | 0.639053254                                      | 47                          |
| 159           | 27                            | 11                                           | 27                                            | A159P;A159V                                                                    | Missense_Mutation                                                                | 0.006390533                                       | 0.639053254                                      | 48                          |
| 194           | 26                            | 15                                           | 26                                            | L194Efs*51;L194F;L194H;L194P;L194Pfs*13;L194R                                  | Frame_Shift_Del;Missense_Mutation                                                | 0.006153846                                       | 0.615384615                                      | 49                          |
| 152           | 25                            | 11                                           | 25                                            | P152Afs*14;P152L;P152Lfs*18;P152Q;P152Rfs*18;P152S;P152Wfs*10                  | Frame_Shift_Del;Missense_Mutation                                                | 0.00591716                                        | 0.591715976                                      | 50                          |
| 261           | 24                            | 11                                           | 24                                            | S261Vfs*84;X261_splice                                                         | Frame_Shift_Del;Splice_Region;Splice_Site                                        | 0.005680473                                       | 0.568047337                                      | 51                          |
| 271           | 24                            | 12                                           | 24                                            | E271*;E271K;E271Q;E271V                                                        | Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation                              | 0.005680473                                       | 0.568047337                                      | 52                          |
| 236           | 22                            | 8                                            | 22                                            | Y236*;Y236C;Y236D;Y236H;Y236del                                                | In_Frame_Del;Missense_Mutation;Nonsense_Mutation                                 | 0.005207101                                       | 0.520710059                                      | 53                          |
| 110           | 22                            | 10                                           | 22                                            | R110C;R110L;R110P;R110Sfs*14;R110Vfs*13;R110Wfs*12;R110_G112del;R110del        | Frame_Shift_Del;Frame_Shift_Ins;In_Frame_Del;Missense_Mutation                   | 0.005207101                                       | 0.520710059                                      | 54                          |
| 294           | 22                            | 10                                           | 22                                            | E294*;E294Sfs*51                                                               | Frame_Shift_Del;Nonsense_Mutation                                                | 0.005207101                                       | 0.520710059                                      | 55                          |
| 136           | 21                            | 11                                           | 21                                            | Q136*;Q136E;Q136H;Q136P;Q136_C141del;Q136del                                   | In_Frame_Del;Missense_Mutation;Nonsense_Mutation                                 | 0.004970412                                       | 0.49704142                                       | 56                          |
| 216           | 21                            | 12                                           | 21                                            | V216E;V216G;V216L;V216M                                                        | Missense_Mutation                                                                | 0.004970412                                       | 0.49704142                                       | 57                          |
| 239           | 20                            | 14                                           | 20                                            | N239*;N239D;N239Qfs*24;N239S;N239_S240del;N239_S241del                         | Frame_Shift_Del;Frame_Shift_Ins;In_Frame_Del;Missense_Mutation                   | 0.004733728                                       | 0.473372781                                      | 58                          |
| 298           | 19                            | 7                                            | 19                                            | E298*                                                                          | Nonsense_Mutation                                                                | 0.004497041                                       | 0.449704142                                      | 59                          |
| 127           | 19                            | 10                                           | 19                                            | S127C;S127F;S127P;S127T;S127Y;S127Yfs*43                                       | Frame_Shift_Del;Missense_Mutation                                                | 0.004497041                                       | 0.449704142                                      | 60                          |
| 161           | 19                            | 10                                           | 18                                            | A161D;A161Gfs*3;A161P;A161S;A161T;A161V;A161_I162del                           | Frame_Shift_Del;In_Frame_Del;Missense_Mutation                                   | 0.004497041                                       | 0.449704142                                      | 61                          |
| 214           | 19                            | 13                                           | 19                                            | H214L;H214Qfs*33;H214R                                                         | Frame_Shift_Del;Missense_Mutation                                                | 0.004497041                                       | 0.449704142                                      | 62                          |
| 215           | 18                            | 14                                           | 18                                            | S215G;S215I;S215Kfs*7;S215N;S215R                                              | Frame_Shift_Ins;Missense_Mutation                                                | 0.004260355                                       | 0.426035503                                      | 63                          |
| 274           | 18                            | 10                                           | 18                                            | V274D;V274F;V274G;V274L;V274_C275dup;V274_G279del;V274dup                      | In_Frame_Del;In_Frame_Ins;Missense_Mutation                                      | 0.004260355                                       | 0.426035503                                      | 64                          |
| 154           | 18                            | 8                                            | 18                                            | G154Afs*16;G154Rfs*27;G154V;G154Wfs*10                                         | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation                                | 0.004260355                                       | 0.426035503                                      | 65                          |
| 144           | 17                            | 11                                           | 17                                            | Q144*;Q144Afs*16;Q144Gfs*24;Q144H;Q144L;Q144Lfs*5;Q144P;Q144Tfs*5;Q144_L145dup | Frame_Shift_Del;Frame_Shift_Ins;In_Frame_Ins;Missense_Mutation;Nonsense_Mutation | 0.004023669                                       | 0.402366864                                      | 66                          |
| 33            | 17                            | 12                                           | 17                                            | S33Ffs*10;X33_splice                                                           | Frame_Shift_Ins;Splice_Site                                                      | 0.004023669                                       | 0.402366864                                      | 67                          |
| 267           | 17                            | 9                                            | 17                                            | R267G;R267L;R267P;R267Q;R267W                                                  | Missense_Mutation                                                                | 0.004023669                                       | 0.402366864                                      | 68                          |
| 317           | 17                            | 12                                           | 17                                            | Q317*;Q317Afs*19;Q317Pfs*20                                                    | Frame_Shift_Del;Frame_Shift_Ins;Nonsense_Mutation                                | 0.004023669                                       | 0.402366864                                      | 69                          |
| 17            | 16                            | 13                                           | 16                                            | E171*;E171Gfs*3;E171Gfs*4;E171K;E171Lfs*2;E1                                   | Frame_Shift_Del;Frame_Shift_Ins;Missense_M                                       | 0.003                                             | 0.378                                            | 70                          |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes                                                                                    | mutation types                                                       | muta<br>tion<br>fre<br>qu<br>ency<br>frac<br>tion | muta<br>tion<br>fre<br>qu<br>ency<br>per<br>cent | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 1             |                               |                                              |                                               | 71Rfs*3                                                                                            | utation;Nonsense_Mutation                                            | 7869<br>82                                        | 6982<br>25                                       |                             |
| 15<br>5       | 16                            | 9                                            | 16                                            | T155I;T155N;T155P;T155_A161del;T155_R156del;<br>T155_R156delinsN;T155_R158del;T155_R158delins<br>C | In_Frame_Del;Missense_Mutation                                       | 0.003<br>7869<br>82                               | 0.378<br>6982<br>25                              | 71                          |
| 25<br>8       | 15                            | 10                                           | 15                                            | E258*;E258A;E258D;E258G;E258K;E258Q;E258Qf<br>s*3;X258_splice                                      | Frame_Shift_Del;Missense_Mutation;Nonsense<br>_Mutation;Splice_Site  | 0.003<br>5502<br>96                               | 0.355<br>0295<br>86                              | 72                          |
| 91            | 15                            | 12                                           | 15                                            | W91*                                                                                               | Nonsense_Mutation                                                    | 0.003<br>5502<br>96                               | 0.355<br>0295<br>86                              | 73                          |
| 14<br>6       | 15                            | 10                                           | 15                                            | W146*;W146S;W146Vfs*3                                                                              | Frame_Shift_Ins;Missense_Mutation;Nonsense<br>_Mutation              | 0.003<br>5502<br>96                               | 0.355<br>0295<br>86                              | 74                          |
| 17<br>7       | 15                            | 8                                            | 15                                            | P177H;P177L;P177R;P177S;P177T;P177_C182del;P<br>177_H178del                                        | In_Frame_Del;Missense_Mutation                                       | 0.003<br>5502<br>96                               | 0.355<br>0295<br>86                              | 75                          |
| 20<br>4       | 15                            | 8                                            | 15                                            | E204*;E204D;E204Q;E204Sfs*43;E204Vfs*4                                                             | Frame_Shift_Del;Missense_Mutation;Nonsense<br>_Mutation              | 0.003<br>5502<br>96                               | 0.355<br>0295<br>86                              | 76                          |
| 10<br>5       | 15                            | 6                                            | 15                                            | G105C;G105D;G105R;G105S;G105V                                                                      | Missense_Mutation                                                    | 0.003<br>5502<br>96                               | 0.355<br>0295<br>86                              | 77                          |
| 20<br>9       | 15                            | 10                                           | 15                                            | R209*;R209Hfs*5;R209I;R209K;R209Kfs*6                                                              | Frame_Shift_Del;Missense_Mutation;Nonsense<br>_Mutation              | 0.003<br>5502<br>96                               | 0.355<br>0295<br>86                              | 78                          |
| 25<br>0       | 15                            | 12                                           | 15                                            | P250Hfs*13;P250L;P250R;P250_I251delinsL                                                            | Frame_Shift_Del;In_Frame_Del;Missense_Mut<br>ation                   | 0.003<br>5502<br>96                               | 0.355<br>0295<br>86                              | 79                          |
| 33<br>2       | 14                            | 11                                           | 14                                            | I332F;X332_splice                                                                                  | Missense_Mutation;Splice_Site                                        | 0.003<br>3136<br>09                               | 0.331<br>3609<br>47                              | 80                          |
| 17<br>8       | 14                            | 11                                           | 14                                            | H178D;H178Pfs*3;H178Pfs*70;H178Q;H178Qfs*3;<br>H178Sfs*69;H178Tfs*69;H178_S183del                  | Frame_Shift_Del;Frame_Shift_Ins;In_Frame_D<br>el;Missense_Mutation   | 0.003<br>3136<br>09                               | 0.331<br>3609<br>47                              | 81                          |
| 27<br>0       | 13                            | 7                                            | 13                                            | F270C;F270I;F270L;F270Lfs*2;F270S;F270V                                                            | Frame_Shift_Ins;Missense_Mutation                                    | 0.003<br>0769<br>23                               | 0.307<br>6923<br>08                              | 82                          |
| 24<br>6       | 13                            | 11                                           | 13                                            | M246I;M246R;M246T;M246V                                                                            | Missense_Mutation                                                    | 0.003<br>0769<br>23                               | 0.307<br>6923<br>08                              | 83                          |
| 27            | 13                            | 11                                           | 13                                            | P27L;P27Lfs*17;P27S                                                                                | Frame_Shift_Del;Missense_Mutation                                    | 0.003<br>0769<br>23                               | 0.307<br>6923<br>08                              | 84                          |
| 15<br>6       | 13                            | 10                                           | 13                                            | R156Afs*12;R156C;R156G;R156H;R156Hfs*26;R15<br>6P;R156Pfs*13;R156del                               | Frame_Shift_Del;Frame_Shift_Ins;In_Frame_D<br>el;Missense_Mutation   | 0.003<br>0769<br>23                               | 0.307<br>6923<br>08                              | 85                          |
| 19<br>8       | 13                            | 8                                            | 13                                            | E198*;E198Gfs*11                                                                                   | Frame_Shift_Ins;Nonsense_Mutation                                    | 0.003<br>0769<br>23                               | 0.307<br>6923<br>08                              | 86                          |
| 27<br>7       | 13                            | 7                                            | 13                                            | C277*;C277F;C277G;C277Vfs*68;C277W;C277Y;C<br>277dup                                               | Frame_Shift_Del;In_Frame_Ins;Missense_Muta<br>tion;Nonsense_Mutation | 0.003<br>0769<br>23                               | 0.307<br>6923<br>08                              | 87                          |
| 13<br>0       | 12                            | 9                                            | 12                                            | L130Cfs*20;L130F;L130P;L130R;L130V                                                                 | Frame_Shift_Ins;Missense_Mutation                                    | 0.002<br>8402<br>37                               | 0.284<br>0236<br>69                              | 88                          |
| 14<br>3       | 12                            | 9                                            | 12                                            | V143A;V143Afs*29;V143E;V143G;V143M;V143Rf<br>s*18                                                  | Frame_Shift_Del;Frame_Shift_Ins;Missense_M<br>utation                | 0.002<br>8402<br>37                               | 0.284<br>0236<br>69                              | 89                          |
| 35            | 12                            | 7                                            | 12                                            | L35Cfs*9;L35Ffs*7;L35Ffs*8;L35Pfs*10                                                               | Frame_Shift_Del;Frame_Shift_Ins                                      | 0.002<br>8402<br>37                               | 0.284<br>0236<br>69                              | 90                          |
| 12<br>2       | 12                            | 7                                            | 12                                            | V122Cfs*27;V122Dfs*26;V122Lfs*25                                                                   | Frame_Shift_Del;Frame_Shift_Ins                                      | 0.002<br>8402<br>37                               | 0.284<br>0236<br>69                              | 91                          |
| 25<br>9       | 12                            | 8                                            | 12                                            | D259Efs*86;D259N;D259V;D259Y                                                                       | Frame_Shift_Del;Missense_Mutation                                    | 0.002<br>8402<br>37                               | 0.284<br>0236<br>69                              | 92                          |
| 25<br>5       | 12                            | 7                                            | 12                                            | I255F;I255N;I255S;I255T;I255del                                                                    | In_Frame_Del;Missense_Mutation                                       | 0.002<br>8402                                     | 0.284<br>0236                                    | 93                          |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes                                                              | mutation types                                      | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ran<br>k |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 26<br>2       | 11                            | 8                                            | 11                                            | G262V;G262Vfs*83;G262del                                                     | Frame_Shift_Del;In_Frame_Del;Missense_Mutation      | 37<br>0.002<br>6035<br>5                          | 69<br>0.260<br>3550<br>3                         | 94                          |
| 32            | 11                            | 6                                            | 11                                            | L32Cfs*12;X32_splice                                                         | Frame_Shift_Del;Splice_Region;Splice_Site           | 0.002<br>6035<br>5                                | 0.260<br>3550<br>3                               | 95                          |
| 13<br>9       | 11                            | 8                                            | 11                                            | K139Afs*5;K139Cfs*6;K139N;K139Nfs*9;K139Rfs*31;K139_P142del;K139_P142delinsT | Frame_Shift_Del;In_Frame_Del;Missense_Mutation      | 0.002<br>6035<br>5                                | 0.260<br>3550<br>3                               | 96                          |
| 19<br>1       | 11                            | 8                                            | 11                                            | P191L;P191Qfs*51;P191del                                                     | Frame_Shift_Del;In_Frame_Del;Missense_Mutation      | 0.002<br>6035<br>5                                | 0.260<br>3550<br>3                               | 97                          |
| 28<br>3       | 10                            | 5                                            | 10                                            | R283Afs*62;R283H;R283P                                                       | Frame_Shift_Del;Missense_Mutation                   | 0.002<br>3668<br>64                               | 0.236<br>6863<br>91                              | 98                          |
| 73            | 10                            | 7                                            | 10                                            | V73Rfs*76;V73Wfs*50                                                          | Frame_Shift_Del;Frame_Shift_Ins                     | 0.002<br>3668<br>64                               | 0.236<br>6863<br>91                              | 99                          |
| 16<br>7       | 10                            | 9                                            | 10                                            | Q167*;Q167Hfs*3;Q167Tfs*14                                                   | Frame_Shift_Del;Frame_Shift_Ins;Nonsense_Mutation   | 0.002<br>3668<br>64                               | 0.236<br>6863<br>91                              | 10<br>0                     |
| 13<br>1       | 10                            | 8                                            | 10                                            | N131I;N131Kfs*39;N131Qfs*17;N131S;N131Tfs*39;N131del                         | Frame_Shift_Del;In_Frame_Del;Missense_Mutation      | 0.002<br>3668<br>64                               | 0.236<br>6863<br>91                              | 10<br>1                     |
| 21<br>1       | 10                            | 6                                            | 10                                            | T211A;T211Ffs*4;T211I;T211Lfs*36;T211_H214del                                | Frame_Shift_Del;In_Frame_Del;Missense_Mutation      | 0.002<br>3668<br>64                               | 0.236<br>6863<br>91                              | 10<br>2                     |
| 13<br>8       | 10                            | 7                                            | 10                                            | A138Cfs*27;A138T;A138V;A138_L145del;A138_P142del                             | Frame_Shift_Del;In_Frame_Del;Missense_Mutation      | 0.002<br>3668<br>64                               | 0.236<br>6863<br>91                              | 10<br>3                     |
| 27<br>9       | 9                             | 7                                            | 9                                             | G279E;G279Pfs*69;G279R                                                       | Frame_Shift_Ins;Missense_Mutation                   | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 10<br>4                     |
| 28<br>7       | 9                             | 5                                            | 9                                             | E287*;E287D;E287Pfs*9;E287Q                                                  | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 10<br>5                     |
| 26<br>5       | 9                             | 7                                            | 9                                             | L265P;L265R;L265Tfs*7                                                        | Frame_Shift_Ins;Missense_Mutation                   | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 10<br>6                     |
| 27<br>6       | 9                             | 8                                            | 9                                             | A276D;A276G;A276Lfs*29;A276Lfs*31;A276P                                      | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation   | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 10<br>7                     |
| 23<br>2       | 9                             | 8                                            | 9                                             | I232F;I232N;I232S;I232T                                                      | Missense_Mutation                                   | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 10<br>8                     |
| 90            | 9                             | 8                                            | 9                                             | S90Ffs*53;S90Pfs*33;S90Pfs*34;S90Vfs*55                                      | Frame_Shift_Del;Frame_Shift_Ins                     | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 10<br>9                     |
| 19<br>9       | 9                             | 7                                            | 9                                             | G199*;G199Efs*48;G199Ifs*47;G199V                                            | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 11<br>0                     |
| 10<br>9       | 9                             | 5                                            | 9                                             | F109C;F109Lfs*36;F109S;F109V                                                 | Frame_Shift_Del;Missense_Mutation                   | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 11<br>1                     |
| 11<br>3       | 9                             | 7                                            | 9                                             | F113C;F113L;F113V;F113del                                                    | In_Frame_Del;Missense_Mutation                      | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 11<br>2                     |
| 18<br>3       | 9                             | 6                                            | 9                                             | S183*                                                                        | Nonsense_Mutation                                   | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 11<br>3                     |
| 33<br>6       | 9                             | 5                                            | 9                                             | E336*;E336Afs*10;E336Sfs*9;E336_R337del                                      | Frame_Shift_Del;In_Frame_Del;Nonsense_Mutation      | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 11<br>4                     |
| 33<br>9       | 9                             | 6                                            | 9                                             | E339*;E339Afs*8;E339Rfs*6                                                    | Frame_Shift_Del;Frame_Shift_Ins;Nonsense_Mutation   | 0.002<br>1301<br>78                               | 0.213<br>0177<br>51                              | 11<br>5                     |
| 17<br>2       | 8                             | 5                                            | 8                                             | V172D;V172F;V172G;V172Lfs*2                                                  | Frame_Shift_Del;Missense_Mutation                   | 0.001<br>8934<br>91                               | 0.189<br>3491<br>12                              | 11<br>6                     |



| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes                                                      | mutation types                                                              | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 190           | 8                             | 5                                            | 8                                             | P190L;P190R                                                          | Missense_Mutation                                                           | 0.001893491                                       | 0.18934912                                       | 117                         |
| 53            | 8                             | 8                                            | 8                                             | W53*;W53Cfs*4;W53Mfs*4                                               | Frame_Shift_Ins;Nonsense_Mutation                                           | 0.001893491                                       | 0.18934912                                       | 118                         |
| 343           | 7                             | 4                                            | 7                                             | E343*;E343Gfs*2                                                      | Frame_Shift_Del;Nonsense_Mutation                                           | 0.001656805                                       | 0.165680473                                      | 119                         |
| 181           | 7                             | 7                                            | 7                                             | R181C;R181H;R181P                                                    | Missense_Mutation                                                           | 0.001656805                                       | 0.165680473                                      | 120                         |
| 162           | 7                             | 5                                            | 7                                             | I162F;I162Hfs*12;I162N;I162Tfs*8;I162Wfs*10;I162_Y163delinsN;I162dup | Frame_Shift_Del;Frame_Shift_Ins;In_Frame_Del;In_Frame_Ins;Missense_Mutation | 0.001656805                                       | 0.165680473                                      | 121                         |
| 334           | 7                             | 4                                            | 7                                             | G334V;G334W                                                          | Missense_Mutation                                                           | 0.001656805                                       | 0.165680473                                      | 122                         |
| 197           | 7                             | 7                                            | 7                                             | V197G;V197Ifs*42;V197L;V197M                                         | Frame_Shift_Del;Missense_Mutation                                           | 0.001656805                                       | 0.165680473                                      | 123                         |
| 134           | 7                             | 6                                            | 7                                             | F134L;F134V                                                          | Missense_Mutation                                                           | 0.001656805                                       | 0.165680473                                      | 124                         |
| 111           | 7                             | 5                                            | 7                                             | L111P;L111Q;L111R                                                    | Missense_Mutation                                                           | 0.001656805                                       | 0.165680473                                      | 125                         |
| 94            | 7                             | 6                                            | 7                                             | S94*;S94Cfs*30                                                       | Frame_Shift_Ins;Nonsense_Mutation                                           | 0.001656805                                       | 0.165680473                                      | 126                         |
| 256           | 7                             | 6                                            | 7                                             | T256Hfs*8;T256I;T256Ifs*90;T256Nfs*8;T256del                         | Frame_Shift_Ins;In_Frame_Del;Missense_Mutation                              | 0.001656805                                       | 0.165680473                                      | 127                         |
| 252           | 7                             | 5                                            | 7                                             | L252P;L252Sfs*93;L252_I254del;L252_T253delinsP;L252del               | Frame_Shift_Del;In_Frame_Del;Missense_Mutation                              | 0.001656805                                       | 0.165680473                                      | 128                         |
| 104           | 7                             | 6                                            | 7                                             | Q104*;Q104H;Q104Tfs*20                                               | Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation                         | 0.001656805                                       | 0.165680473                                      | 129                         |
| 229           | 7                             | 7                                            | 7                                             | C229*;C229Lfs*18;C229Yfs*10;C229_I232del                             | Frame_Shift_Del;In_Frame_Del;Nonsense_Mutation                              | 0.001656805                                       | 0.165680473                                      | 130                         |
| 180           | 6                             | 6                                            | 6                                             | E180*;E180K;E180_D184del;E180_S185del                                | In_Frame_Del;Missense_Mutation;Nonsense_Mutation                            | 0.001420118                                       | 0.142011834                                      | 131                         |
| 253           | 6                             | 5                                            | 5                                             | T253A;T253Hfs*11;T253I;T253N;T253dup                                 | Frame_Shift_Ins;In_Frame_Ins;Missense_Mutation                              | 0.001420118                                       | 0.142011834                                      | 132                         |
| 120           | 6                             | 5                                            | 6                                             | K120E;K120Sfs*3;K120Tfs*2                                            | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                         | 0.001420118                                       | 0.142011834                                      | 133                         |
| 164           | 6                             | 6                                            | 6                                             | K164E;K164Sfs*5;K164Sfs*6                                            | Frame_Shift_Del;Missense_Mutation                                           | 0.001420118                                       | 0.142011834                                      | 134                         |
| 124           | 6                             | 5                                            | 6                                             | C124*;C124Afs*46;C124G;C124Wfs*25                                    | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation         | 0.001420118                                       | 0.142011834                                      | 135                         |
| 128           | 6                             | 5                                            | 6                                             | P128Lfs*42;P128del                                                   | Frame_Shift_Del;In_Frame_Del                                                | 0.001420118                                       | 0.142011834                                      | 136                         |
| 65            | 6                             | 3                                            | 6                                             | R65*;R65Efs*58;R65Qfs*84                                             | Frame_Shift_Del;Frame_Shift_Ins;Nonsense_Mutation                           | 0.001420118                                       | 0.142011834                                      | 137                         |
| 68            | 6                             | 4                                            | 6                                             | E68*                                                                 | Nonsense_Mutation                                                           | 0.001420118                                       | 0.142011834                                      | 138                         |
| 257           | 6                             | 5                                            | 6                                             | L257P;L257Pfs*7;L257Q;L257R                                          | Frame_Shift_Ins;Missense_Mutation                                           | 0.001420118                                       | 0.142011834                                      | 139                         |
| 16            | 6                             | 6                                            | 6                                             | H168Cfs*8;H168L;H168P;H168R                                          | Frame_Shift_Del;Missense_Mutation                                           | 0.001                                             | 0.142                                            | 14                          |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes                                | mutation types                                                 | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 8             |                               |                                              |                                               |                                                |                                                                | 4201<br>18                                        | 0118<br>34                                       | 0                           |
| 21<br>8       | 6                             | 5                                            | 6                                             | V218Cfs*29;V218E;V218G;V218del                 | Frame_Shift_Del;In_Frame_Del;Missense_Mutation                 | 0.001<br>4201<br>18                               | 0.142<br>0118<br>34                              | 14<br>1                     |
| 22<br>1       | 6                             | 6                                            | 6                                             | E221*;E221Afs*2                                | Frame_Shift_Del;Nonsense_Mutation                              | 0.001<br>4201<br>18                               | 0.142<br>0118<br>34                              | 14<br>2                     |
| 24<br>0       | 6                             | 5                                            | 6                                             | S240G;S240Gfs*20;S240Kfs*24;S240R;S240_C242del | Frame_Shift_Del;Frame_Shift_Ins;In_Frame_Del;Missense_Mutation | 0.001<br>4201<br>18                               | 0.142<br>0118<br>34                              | 14<br>3                     |
| 25<br>1       | 6                             | 5                                            | 6                                             | I251F;I251N;I251S;I251Sfs*94                   | Frame_Shift_Del;Missense_Mutation                              | 0.001<br>4201<br>18                               | 0.142<br>0118<br>34                              | 14<br>4                     |
| 56            | 6                             | 5                                            | 6                                             | E56*;E56Kfs*67                                 | Frame_Shift_Del;Nonsense_Mutation                              | 0.001<br>4201<br>18                               | 0.142<br>0118<br>34                              | 14<br>5                     |
| 30<br>1       | 6                             | 5                                            | 6                                             | P301Qfs*44;P301T                               | Frame_Shift_Del;Missense_Mutation                              | 0.001<br>4201<br>18                               | 0.142<br>0118<br>34                              | 14<br>6                     |
| 15<br>3       | 5                             | 4                                            | 5                                             | P153Afs*28;P153Rfs*18                          | Frame_Shift_Ins                                                | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 14<br>7                     |
| 33<br>5       | 5                             | 4                                            | 5                                             | R335Lfs*10;R335Qfs*2;R335Vfs*10                | Frame_Shift_Del;Frame_Shift_Ins                                | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 14<br>8                     |
| 10<br>6       | 5                             | 4                                            | 5                                             | S106R                                          | Missense_Mutation                                              | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 14<br>9                     |
| 10<br>0       | 5                             | 4                                            | 5                                             | Q100*;Q100Gfs*37                               | Frame_Shift_Del;Nonsense_Mutation                              | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>0                     |
| 22<br>8       | 5                             | 4                                            | 5                                             | D228*;D228E;D228N;D228Vfs*18                   | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation              | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>1                     |
| 30<br>2       | 5                             | 5                                            | 5                                             | G302Afs*31;G302Efs*3;G302Rfs*4                 | Frame_Shift_Del;Frame_Shift_Ins                                | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>2                     |
| 10<br>3       | 5                             | 4                                            | 5                                             | Y103*;Y103Lfs*46;Y103Tfs*20                    | Frame_Shift_Del;Frame_Shift_Ins;Nonsense_Mutation              | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>3                     |
| 10<br>8       | 5                             | 3                                            | 5                                             | G108Ffs*40;G108S;G108Vfs*15                    | Frame_Shift_Del;Missense_Mutation                              | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>4                     |
| 30<br>5       | 5                             | 4                                            | 5                                             | K305*;K305Efs*39;X305_splice                   | Frame_Shift_Del;Nonsense_Mutation;Splice_Site                  | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>5                     |
| 38<br>2       | 5                             | 3                                            | 5                                             | K382Nfs*40                                     | Frame_Shift_Del                                                | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>6                     |
| 34<br>9       | 5                             | 4                                            | 5                                             | E349*                                          | Nonsense_Mutation                                              | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>7                     |
| 14<br>5       | 5                             | 4                                            | 5                                             | L145P;L145Q;L145R                              | Missense_Mutation                                              | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>8                     |
| 38            | 5                             | 5                                            | 5                                             | Q38*;Q38Kfs*6                                  | Frame_Shift_Del;Nonsense_Mutation                              | 0.001<br>1834<br>32                               | 0.118<br>3431<br>95                              | 15<br>9                     |
| 21<br>2       | 4                             | 4                                            | 4                                             | F212Sfs*3;F212Yfs*34                           | Frame_Shift_Del                                                | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 16<br>0                     |
| 20<br>3       | 4                             | 4                                            | 4                                             | V203Gfs*44;V203Gfs*5;V203L;V203Wfs*44          | Frame_Shift_Del;Missense_Mutation                              | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 16<br>1                     |
| 10<br>7       | 4                             | 3                                            | 4                                             | Y107Cfs*16;Y107D                               | Frame_Shift_Del;Missense_Mutation                              | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 16<br>2                     |
| 62            | 4                             | 4                                            | 4                                             | E62*;E62K;E62Kfs*61                            | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation            | 0.000<br>9467                                     | 0.094<br>6745                                    | 16<br>3                     |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes                          | mutation types                                                      | muta<br>tion<br>fre<br>qu<br>ency<br>frac<br>tion | muta<br>tion<br>fre<br>qu<br>ency<br>per<br>cent | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 268           | 4                             | 4                                            | 4                                             | N268*;N268Kfs*77;N268Tfs*77;N268_R273del | Frame_Shift_Del;In_Frame_Del                                        | 0.000<br>9467<br>46                               | 56<br>0.094<br>6745<br>56                        | 16<br>4                     |
| 97            | 4                             | 4                                            | 4                                             | V97D;V97Efs*48;V97Sfs*26                 | Frame_Shift_Del;Missense_Mutation                                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 16<br>5                     |
| 348           | 4                             | 3                                            | 4                                             | L348*;L348F;L348S;L348Wfs*22             | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                 | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 16<br>6                     |
| 174           | 4                             | 4                                            | 4                                             | R174Sfs*73;R174W                         | Frame_Shift_Del;Missense_Mutation                                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 16<br>7                     |
| 166           | 4                             | 4                                            | 4                                             | S166*                                    | Nonsense_Mutation                                                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 16<br>8                     |
| 333           | 4                             | 4                                            | 4                                             | R333C;R333Vfs*12                         | Frame_Shift_Del;Missense_Mutation                                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 16<br>9                     |
| 208           | 4                             | 4                                            | 4                                             | D208G;D208N;D208V                        | Missense_Mutation                                                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 17<br>0                     |
| 48            | 4                             | 4                                            | 4                                             | D48Gfs*4;D48N;D48Nfs*72;D48Tfs*75        | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 17<br>1                     |
| 51            | 4                             | 3                                            | 4                                             | E51*;E51Gfs*6                            | Frame_Shift_Ins;Nonsense_Mutation                                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 17<br>2                     |
| 330           | 4                             | 3                                            | 4                                             | L330Ffs*15;L330I;L330R                   | Frame_Shift_Del;Missense_Mutation                                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 17<br>3                     |
| 367           | 4                             | 3                                            | 4                                             | X367_splice                              | Splice_Site                                                         | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 17<br>4                     |
| 319           | 4                             | 4                                            | 4                                             | K319*;K319Afs*19;K319N;K319Rfs*26        | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation;Nonsense_Mutation | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 17<br>5                     |
| 223           | 4                             | 4                                            | 4                                             | P223*;P223Rfs*4;X223_splice              | Frame_Shift_Del;Frame_Shift_Ins;Splice_Site                         | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 17<br>6                     |
| 320           | 4                             | 3                                            | 3                                             | K320E;K320Gfs*22;K320Nfs*19;K320Rfs*25   | Frame_Shift_Del;Missense_Mutation                                   | 0.000<br>9467<br>46                               | 0.094<br>6745<br>56                              | 17<br>7                     |
| 25            | 3                             | 3                                            | 3                                             | X25_splice                               | Splice_Site                                                         | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 17<br>8                     |
| 345           | 3                             | 2                                            | 3                                             | N345D;N345Mfs*25;N345Sfs*2               | Frame_Shift_Del;Missense_Mutation;Nonsense_Mutation                 | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 17<br>9                     |
| 341           | 3                             | 3                                            | 3                                             | F341Efs*7;F341L;F341V                    | Frame_Shift_Ins;Missense_Mutation                                   | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 18<br>0                     |
| 64            | 3                             | 3                                            | 3                                             | P64Qfs*84;P64Sfs*59;P64T                 | Frame_Shift_Del;Missense_Mutation                                   | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 18<br>1                     |
| 263           | 3                             | 2                                            | 3                                             | N263Ifs*82;N263Kfs*9;N263Sfs*8           | Frame_Shift_Del;Frame_Shift_Ins                                     | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 18<br>2                     |
| 254           | 3                             | 3                                            | 3                                             | I254S                                    | Missense_Mutation                                                   | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 18<br>3                     |
| 72            | 3                             | 3                                            | 3                                             | P72Rfs*51;P72S                           | Frame_Shift_Del;Missense_Mutation                                   | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 18<br>4                     |
| 313           | 3                             | 3                                            | 3                                             | S313Afs*32;S313Tfs*32                    | Frame_Shift_Del                                                     | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 18<br>5                     |
| 375           | 3                             | 3                                            | 3                                             | Q375*;Q375K                              | Missense_Mutation;Nonsense_Mutation                                 | 0.000<br>7100<br>59                               | 0.071<br>0059<br>17                              | 18<br>6                     |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes                  | mutation types                                    | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ran<br>k |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|----------------------------------|---------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 290           | 3                             | 3                                            | 3                                             | R290C;R290Kfs*53;R290Sfs*56      | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation | 0.000710059                                       | 0.071005917                                      | 187                         |
| 52            | 3                             | 3                                            | 3                                             | Q52*                             | Nonsense_Mutation                                 | 0.000710059                                       | 0.071005917                                      | 188                         |
| 284           | 3                             | 2                                            | 3                                             | T284Hfs*22;T284Kfs*61;T284Qfs*61 | Frame_Shift_Del;Frame_Shift_Ins                   | 0.000710059                                       | 0.071005917                                      | 189                         |
| 292           | 3                             | 3                                            | 3                                             | K292*;K292Gfs*52                 | Frame_Shift_Del;Nonsense_Mutation                 | 0.000710059                                       | 0.071005917                                      | 190                         |
| 77            | 3                             | 2                                            | 3                                             | P77Cfs*73;P77Gfs*65;P77L         | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation | 0.000710059                                       | 0.071005917                                      | 191                         |
| 43            | 3                             | 3                                            | 3                                             | L43*;L43Afs*7                    | Frame_Shift_Del                                   | 0.000710059                                       | 0.071005917                                      | 192                         |
| 118           | 3                             | 3                                            | 3                                             | T118Dfs*31;T118Qfs*5             | Frame_Shift_Del;Frame_Shift_Ins                   | 0.000710059                                       | 0.071005917                                      | 193                         |
| 230           | 3                             | 3                                            | 3                                             | T230Hfs*9;T230P;T230_T231del     | Frame_Shift_Del;In_Frame_Del;Missense_Mutation    | 0.000710059                                       | 0.071005917                                      | 194                         |
| 123           | 3                             | 3                                            | 3                                             | T123Dfs*26;T123Lfs*47;T123Rfs*48 | Frame_Shift_Del;Frame_Shift_Ins                   | 0.000710059                                       | 0.071005917                                      | 195                         |
| 219           | 3                             | 3                                            | 3                                             | P219Afs*3;P219Lfs*2;P219T        | Frame_Shift_Del;Frame_Shift_Ins;Missense_Mutation | 0.000710059                                       | 0.071005917                                      | 196                         |
| 165           | 3                             | 3                                            | 3                                             | Q165*;Q165Afs*6;Q165Hfs*17       | Frame_Shift_Del;Frame_Shift_Ins;Nonsense_Mutation | 0.000710059                                       | 0.071005917                                      | 197                         |
| 149           | 3                             | 3                                            | 3                                             | S149Ffs*32;S149Pfs*21            | Frame_Shift_Del;Frame_Shift_Ins                   | 0.000710059                                       | 0.071005917                                      | 198                         |
| 169           | 3                             | 3                                            | 3                                             | M169I;M169T                      | Missense_Mutation                                 | 0.000710059                                       | 0.071005917                                      | 199                         |
| 189           | 3                             | 3                                            | 3                                             | A189Dfs*14;A189Pfs*58            | Frame_Shift_Del                                   | 0.000710059                                       | 0.071005917                                      | 200                         |
| 142           | 3                             | 3                                            | 3                                             | P142L;P142Lfs*28;P142del         | Frame_Shift_Del;In_Frame_Del;Missense_Mutation    | 0.000710059                                       | 0.071005917                                      | 201                         |
| 147           | 3                             | 3                                            | 3                                             | V147G;V147Lfs*23;V147_D148dup    | Frame_Shift_Del;In_Frame_Ins;Missense_Mutation    | 0.000710059                                       | 0.071005917                                      | 202                         |
| 133           | 3                             | 3                                            | 3                                             | M133K;M133R                      | Missense_Mutation                                 | 0.000710059                                       | 0.071005917                                      | 203                         |
| 201           | 3                             | 2                                            | 3                                             | L201*;L201Ffs*8                  | Frame_Shift_Ins;Nonsense_Mutation                 | 0.000710059                                       | 0.071005917                                      | 204                         |
| 86            | 3                             | 3                                            | 3                                             | A86Cfs*63;A86Pfs*34;A86Vfs*55    | Frame_Shift_Del;Frame_Shift_Ins                   | 0.000710059                                       | 0.071005917                                      | 205                         |
| 329           | 3                             | 2                                            | 3                                             | T329Hfs*8;T329Rfs*14             | Frame_Shift_Del;Frame_Shift_Ins                   | 0.000710059                                       | 0.071005917                                      | 206                         |
| 327           | 3                             | 3                                            | 3                                             | Y327*                            | Nonsense_Mutation                                 | 0.000710059                                       | 0.071005917                                      | 207                         |
| 351           | 3                             | 3                                            | 3                                             | K351*;K351E                      | Missense_Mutation;Nonsense_Mutation               | 0.000710059                                       | 0.071005917                                      | 208                         |
| 58            | 2                             | 1                                            | 2                                             | P58Qfs*65                        | Frame_Shift_Del                                   | 0.000473373                                       | 0.047337278                                      | 209                         |
| 84            | 2                             | 2                                            | 2                                             | A84Pfs*39                        | Frame Shift Del                                   | 0.000                                             | 0.047                                            | 21                          |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes       | mutation types                      | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ran<br>k |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|-----------------------|-------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
|               |                               |                                              |                                               |                       |                                     | 4733<br>73                                        | 3372<br>78                                       | 0                           |
| 22<br>6       | 2                             | 2                                            | 2                                             | G226Afs*21            | Frame_Shift_Del                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>1                     |
| 32<br>1       | 2                             | 2                                            | 2                                             | K321*;K321Efs*16      | Frame_Shift_Ins;Nonsense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>2                     |
| 61            | 2                             | 1                                            | 2                                             | D61*;D61G             | Frame_Shift_Del;Missense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>3                     |
| 31<br>5       | 2                             | 2                                            | 2                                             | S315C;S315Lfs*30      | Frame_Shift_Del;Missense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>4                     |
| 32<br>8       | 2                             | 2                                            | 2                                             | F328Sfs*17            | Frame_Shift_Del                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>5                     |
| 57            | 2                             | 2                                            | 2                                             | D57Kfs*67;D57N        | Frame_Shift_Ins;Missense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>6                     |
| 82            | 2                             | 2                                            | 2                                             | P82L;P82Rfs*41        | Frame_Shift_Del;Missense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>7                     |
| 29<br>7       | 2                             | 2                                            | 2                                             | H297Pfs*48;H297Tfs*48 | Frame_Shift_Del                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>8                     |
| 79            | 2                             | 2                                            | 2                                             | A79Pfs*70;A79V        | Frame_Shift_Ins;Missense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 21<br>9                     |
| 75            | 2                             | 1                                            | 2                                             | P75Lfs*48             | Frame_Shift_Del                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>0                     |
| 18<br>8       | 2                             | 2                                            | 2                                             | L188Gfs*20;L188Tfs*21 | Frame_Shift_Del;Frame_Shift_Ins     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>1                     |
| 34<br>6       | 2                             | 2                                            | 2                                             | E346*                 | Frame_Shift_Ins;Nonsense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>2                     |
| 33<br>8       | 2                             | 2                                            | 2                                             | F338I;F338Lfs*7       | Frame_Shift_Del;Missense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>3                     |
| 11            | 2                             | 2                                            | 2                                             | E11*;E11K             | Missense_Mutation;Nonsense_Mutation | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>4                     |
| 20            | 2                             | 2                                            | 2                                             | S20*;S20Qfs*24        | Frame_Shift_Del;Nonsense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>5                     |
| 14<br>8       | 2                             | 2                                            | 2                                             | D148Ifs*22;D148N      | Frame_Shift_Del;Missense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>6                     |
| 13<br>7       | 2                             | 1                                            | 2                                             | L137Pfs*32;L137Q      | Frame_Shift_Del;Missense_Mutation   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>7                     |
| 20<br>0       | 2                             | 2                                            | 2                                             | N200Ifs*47            | Frame_Shift_Del                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>8                     |
| 46            | 2                             | 1                                            | 2                                             | S46*;S46Tfs*5         | Frame_Shift_Del                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 22<br>9                     |
| 30<br>8       | 2                             | 2                                            | 2                                             | L308Afs*28;L308Qfs*27 | Frame_Shift_Del                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 23<br>0                     |
| 28<br>8       | 2                             | 2                                            | 2                                             | N288Efs*18;N288Kfs*59 | Frame_Shift_Ins                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 23<br>1                     |
| 44            | 2                             | 2                                            | 2                                             | M44Ifs*79;M44Tfs*75   | Frame_Shift_Del                     | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 23<br>2                     |
| 12<br>1       | 2                             | 2                                            | 2                                             | S121Cfs*27;S121Y      | Frame_Shift_Del;Missense_Mutation   | 0.000<br>4733                                     | 0.047<br>3372                                    | 23<br>3                     |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes            | mutation types                    | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|----------------------------|-----------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 92            | 2                             | 2                                            | 2                                             | P92Afs*57;P92S             | Frame_Shift_Ins;Missense_Mutation | 73<br>0.000<br>4733<br>73                         | 78<br>0.047<br>3372<br>78                        | 23<br>4                     |
| 26<br>0       | 2                             | 1                                            | 2                                             | S260Lfs*4;S260_L264dup     | Frame_Shift_Ins;In_Frame_Ins      | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 23<br>5                     |
| 26<br>9       | 2                             | 2                                            | 2                                             | S269Ifs*2;S269_E271delinsK | Frame_Shift_Del;In_Frame_Del      | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 23<br>6                     |
| 20<br>7       | 2                             | 1                                            | 2                                             | D207Ffs*35;D207Mfs*40      | Frame_Shift_Del                   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 23<br>7                     |
| 10<br>2       | 2                             | 2                                            | 2                                             | T102Pfs*21                 | Frame_Shift_Del                   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 23<br>8                     |
| 99            | 2                             | 2                                            | 2                                             | S99Rfs*23                  | Frame_Shift_Del                   | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 23<br>9                     |
| 24<br>7       | 2                             | 2                                            | 2                                             | N247I                      | Missense_Mutation                 | 0.000<br>4733<br>73                               | 0.047<br>3372<br>78                              | 24<br>0                     |
| 18<br>4       | 1                             | 1                                            | 1                                             | D184Afs*62                 | Frame_Shift_Del                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>1                     |
| 1             | 1                             | 1                                            | 1                                             | M1?                        | Translation_Start_Site            | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>2                     |
| 10            | 1                             | 1                                            | 1                                             | X10_splice                 | Splice_Site                       | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>3                     |
| 4             | 1                             | 1                                            | 1                                             | P4L                        | Missense_Mutation                 | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>4                     |
| 30            | 1                             | 1                                            | 1                                             | N30del                     | In_Frame_Del                      | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>5                     |
| 23            | 1                             | 1                                            | 1                                             | W23*                       | Nonsense_Mutation                 | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>6                     |
| 22            | 1                             | 1                                            | 1                                             | L22Yfs*22                  | Frame_Shift_Del                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>7                     |
| 37            | 1                             | 1                                            | 1                                             | S37Vfs*6                   | Frame_Shift_Del                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>8                     |
| 34<br>0       | 1                             | 1                                            | 1                                             | M340Sfs*8                  | Frame_Shift_Ins                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 24<br>9                     |
| 36            | 1                             | 1                                            | 1                                             | P36Wfs*4                   | Frame_Shift_Del                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 25<br>0                     |
| 34<br>7       | 1                             | 1                                            | 1                                             | A347V                      | Missense_Mutation                 | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 25<br>1                     |
| 16<br>0       | 1                             | 1                                            | 1                                             | M160I                      | Missense_Mutation                 | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 25<br>2                     |
| 12<br>9       | 1                             | 1                                            | 1                                             | A129Vfs*20                 | Frame_Shift_Ins                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 25<br>3                     |
| 15<br>0       | 1                             | 1                                            | 1                                             | T150Afs*16                 | Frame_Shift_Del                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 25<br>4                     |
| 18<br>6       | 1                             | 1                                            | 1                                             | D186Vfs*61                 | Frame_Shift_Del                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 25<br>5                     |
| 14<br>0       | 1                             | 1                                            | 1                                             | T140Mfs*28                 | Frame_Shift_Del                   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 25<br>6                     |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes | mutation types    | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ran<br>k |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|-----------------|-------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
| 206           | 1                             | 1                                            | 1                                             | L206Wfs*41      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 257                         |
| 96            | 1                             | 1                                            | 1                                             | S96Ffs*25       | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 258                         |
| 243           | 1                             | 1                                            | 1                                             | M243I           | Missense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 259                         |
| 101           | 1                             | 1                                            | 1                                             | K101*           | Nonsense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 260                         |
| 93            | 1                             | 1                                            | 1                                             | L93Vfs*55       | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 261                         |
| 202           | 1                             | 1                                            | 1                                             | R202_L206del    | In_Frame_Del      | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 262                         |
| 210           | 1                             | 1                                            | 1                                             | N210Tfs*37      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 263                         |
| 217           | 1                             | 1                                            | 1                                             | V217Gfs*31      | Frame_Shift_Ins   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 264                         |
| 114           | 1                             | 1                                            | 1                                             | L114*           | Nonsense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 265                         |
| 115           | 1                             | 1                                            | 1                                             | H115Ifs*8       | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 266                         |
| 227           | 1                             | 1                                            | 1                                             | S227*           | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 267                         |
| 85            | 1                             | 1                                            | 1                                             | P85Lfs*38       | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 268                         |
| 69            | 1                             | 1                                            | 1                                             | A69Gfs*80       | Frame_Shift_Ins   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 269                         |
| 67            | 1                             | 1                                            | 1                                             | P67S            | Missense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 270                         |
| 66            | 1                             | 1                                            | 1                                             | M66Tfs*60       | Frame_Shift_Ins   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 271                         |
| 264           | 1                             | 1                                            | 1                                             | L264Yfs*81      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 272                         |
| 98            | 1                             | 1                                            | 1                                             | P98Afs*51       | Frame_Shift_Ins   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 273                         |
| 303           | 1                             | 1                                            | 1                                             | S303Afs*42      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 274                         |
| 304           | 1                             | 1                                            | 1                                             | T304Ifs*41      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 275                         |
| 40            | 1                             | 1                                            | 1                                             | M40Lfs*7        | Frame_Shift_Ins   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 276                         |
| 291           | 1                             | 1                                            | 1                                             | K291Sfs*51      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 277                         |
| 83            | 1                             | 1                                            | 1                                             | A83Gfs*66       | Frame_Shift_Ins   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 278                         |
| 74            | 1                             | 1                                            | 1                                             | A74T            | Missense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 279                         |
| 76            | 1                             | 1                                            | 1                                             | A76Hfs*47       | Frame Shift Del   | 0.000                                             | 0.023                                            | 28                          |

| co<br>do<br>n | mut<br>atio<br>n<br>cou<br>nt | ca<br>nc<br>er<br>stu<br>dy<br>co<br>un<br>t | uni<br>qu<br>e<br>sam<br>ple<br>co<br>un<br>t | protein changes | mutation types    | muta<br>tion<br>fre<br>qu<br>ency<br>fracti<br>on | muta<br>tion<br>fre<br>qu<br>ency<br>perce<br>nt | hot<br>spo<br>t<br>ra<br>nk |
|---------------|-------------------------------|----------------------------------------------|-----------------------------------------------|-----------------|-------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------|
|               |                               |                                              |                                               |                 |                   | 2366<br>86                                        | 6686<br>39                                       | 0                           |
| 29<br>6       | 1                             | 1                                            | 1                                             | H296Tfs*49      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>1                     |
| 28<br>9       | 1                             | 1                                            | 1                                             | L289F           | Missense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>2                     |
| 31<br>0       | 1                             | 1                                            | 1                                             | N310Tfs*35      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>3                     |
| 29<br>9       | 1                             | 1                                            | 1                                             | L299Afs*7       | Frame_Shift_Ins   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>4                     |
| 49            | 1                             | 1                                            | 1                                             | D49Sfs*69       | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>5                     |
| 47            | 1                             | 1                                            | 1                                             | P47Rfs*76       | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>6                     |
| 42            | 1                             | 1                                            | 1                                             | D42Ifs*2        | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>7                     |
| 32<br>5       | 1                             | 1                                            | 1                                             | G325*           | Nonsense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>8                     |
| 54            | 1                             | 1                                            | 1                                             | F54Sfs*69       | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 28<br>9                     |
| 32<br>2       | 1                             | 1                                            | 1                                             | P322Hfs*23      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>0                     |
| 31<br>6       | 1                             | 1                                            | 1                                             | P316Sfs*21      | Frame_Shift_Ins   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>1                     |
| 31<br>8       | 1                             | 1                                            | 1                                             | P318Tfs*15      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>2                     |
| 35<br>5       | 1                             | 1                                            | 1                                             | A355T           | Missense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>3                     |
| 35<br>4       | 1                             | 1                                            | 1                                             | Q354*           | Nonsense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>4                     |
| 37<br>6       | 1                             | 1                                            | 1                                             | S376C           | Missense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>5                     |
| 37<br>9       | 1                             | 1                                            | 1                                             | R379C           | Missense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>6                     |
| 38<br>3       | 1                             | 1                                            | 1                                             | L383Cfs*38      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>7                     |
| 38<br>5       | 1                             | 1                                            | 1                                             | F385L           | Missense_Mutation | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>8                     |
| 39<br>0       | 1                             | 1                                            | 1                                             | P390Qfs*32      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 29<br>9                     |
| 39<br>2       | 1                             | 1                                            | 1                                             | S392Tfs*76      | Frame_Shift_Del   | 0.000<br>2366<br>86                               | 0.023<br>6686<br>39                              | 30<br>0                     |



**Table S5.** Structural class summary.

| hotspot class          | structural annotation            | n positions | total mutation count | mean mutation count | mean conservation | median conservation | mean entropy |
|------------------------|----------------------------------|-------------|----------------------|---------------------|-------------------|---------------------|--------------|
| canonical hotspot      | DNA contact                      | 2           | 493                  | 246.5               | 1                 | 1                   | 0            |
| canonical hotspot      | structural core                  | 4           | 396                  | 99                  | 1                 | 1                   | 0            |
| recurrent noncanonical | DBD recurrent                    | 8           | 529                  | 66.125              | 0.997767857       | 1                   | 0.016154222  |
| recurrent noncanonical | oligomerization domain recurrent | 1           | 59                   | 59                  | 0.821428571       | 0.821428571         | 0.67694187   |

## Supplementary References and Source Data

### Reference list

1. Cerami, E. et al. The cBio Cancer Genomics Portal: An Open Platform for Exploring Multidimensional Cancer Genomics Data. *Cancer Discovery* 2, 401–404 (2012).
2. Gao, J. et al. Integrative Analysis of Complex Cancer Genomics and Clinical Profiles Using the cBioPortal. *Science Signaling* 6, p11 (2013).
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4. Minh, B. Q. et al. IQ-TREE 2: New Models and Efficient Methods for Phylogenetic Inference in the Genomic Era. *Molecular Biology and Evolution* 37, 1530–1534 (2020).
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7. Cock, P. J. A. et al. Biopython: Freely Available Python Tools for Computational Molecular Biology and Bioinformatics. *Bioinformatics* 25, 1422–1423 (2009). doi:10.1093/bioinformatics/btp163.
8. Olivier, M., Hollstein, M. & Hainaut, P. TP53 Mutations in Human Cancers: Origins, Consequences, and Clinical Use. *Cold Spring Harbor Perspectives in Biology* 2, a001008 (2010).
9. Joerger, A. C. & Fersht, A. R. Structural Biology of the Tumor Suppressor p53. *Annual Review of Biochemistry* 77, 557–582 (2008). doi:10.1146/annurev.biochem.77.060806.091238.
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11. Olivier, M. et al. The IARC TP53 Database: New Online Mutation Analysis and Recommendations to Users. *Human Mutation* 19, 607–614 (2002).
12. Liu, J. et al. An Integrated TCGA Pan-Cancer Clinical Data Resource to Drive High-Quality Survival Outcome Analytics. *Cell* 173, 400–416.e11 (2018). doi:10.1016/j.cell.2018.02.052.

### Source databases and reference targets

**Source table.** Source databases, tools and reference targets used in the supplementary analysis.

| Source or tool                          | Role in supplementary analysis                                                                          |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------|
| cBioPortal / TCGA PanCancer             | Human TP53 mutation-frequency data used for codon-level recurrence analyses.                            |
| NCBI Protein database                   | Source of mammalian TP53 protein accession records used for sequence retrieval and curation.            |
| MAFFT                                   | Multiple sequence alignment method used for curated mammalian TP53 sequences.                           |
| IQ-TREE / ModelFinder / UFBoot2         | Maximum-likelihood phylogenetic reconstruction, model selection and bootstrap support assessment.       |
| Biopython / Python analysis environment | Sequence handling, computational parsing and analysis scripting environment referenced by the workflow. |

### Reference target notes

### Reference Targets v2

#### Core citations already selected

- cBioPortal platform: Cerami et al. 2012; Gao et al. 2013
- MAFFT: Katoh and Standley 2013
- IQ-TREE 2: Minh et al. 2020
- ModelFinder: Kalyaanamoorthy et al. 2017
- UFBoot2: Hoang et al. 2018
- Biopython: Cock et al. 2009
- TP53 mutation biology: Olivier, Hollstein and Hainaut 2010
- TP53 structural biology: Joerger and Fersht 2008
- TP53 mutation database/function: Petitjean et al. 2007; Olivier et al. 2002

#### Still needs verification before final submission

- Exact TCGA PanCancer Atlas citation preferred by the journal/cBioPortal study page.
- Exact DOI/page details for Petitjean et al. 2007 and Olivier et al. 2002 if final reference style requires DOI.
- Whether Computational Biology and Chemistry wants numbered references or author-year style.

### Supplied source-data files

**Source-data table.** Descriptions of supplied reference and source-data files.

| File                    | Description                                                              |
|-------------------------|--------------------------------------------------------------------------|
| REFERENCES v1.md        | Reference list used by the supplementary and main manuscript files.      |
| REFERENCE TARGETS v2.md | Reference target audit listing core citation categories and verification |

| File                     | Description                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| TP53_curated_v0.fasta    | notes.<br>Curated FASTA dataset containing 56 mammalian TP53 protein sequences; sequence lengths range from 361 to 408 amino acids. |
| TP53_mammals_v0.iqtree   | IQ-TREE output log for model selection and maximum-likelihood phylogenetic analysis.                                                |
| TP53_mammals_v0.treefile | Newick-format maximum-likelihood treefile containing taxon labels, branch lengths and support annotations.                          |

#### FASTA dataset description

The supplied FASTA file TP53\_curated\_v0.fasta contains 56 curated mammalian TP53 protein sequences. The sequence length range is 361-408 amino acids. Human TP53 residue-level analyses were mapped to 393 human reference positions.

**FASTA source-data summary.** Mammalian order representation in the curated TP53 sequence audit.

| Mammalian order | Sequence count |
|-----------------|----------------|
| Primates        | 8              |
| Rodentia        | 8              |
| Carnivora       | 8              |
| Artiodactyla    | 7              |
| Cetacea         | 5              |
| Chiroptera      | 5              |
| Perissodactyla  | 3              |
| Proboscidea     | 2              |
| Diprotodontia   | 2              |
| Lagomorpha      | 2              |
| Eulipotyphla    | 2              |
| Dasyuromorphia  | 1              |
| Didelphimorphia | 1              |
| Cingulata       | 1              |
| Sirenia         | 1              |

#### IQ-TREE output summary

IQ-TREE 2.2.6 COVID-edition built Dec 8 2023

Input file name: results/phylogeny/TP53\_curated\_v0\_mafft\_for\_iqtree.fasta

Type of analysis: ModelFinder + tree reconstruction + ultrafast bootstrap (1000 replicates)

Random seed number: 383656

Input data: 56 sequences with 464 amino-acid sites

Number of constant sites: 169 (= 36.4224% of all sites)

Number of parsimony informative sites: 221

Number of distinct site patterns: 361

Best-fit model according to BIC: Q.bird+I+G4

*The supplied IQ-TREE file also contains the full model ranking table, citation recommendations, sequence-alignment summary and tree reconstruction log.*

#### Treefile description

The supplied TP53\_mammals\_v0.treefile is a Newick-format maximum-likelihood phylogeny for the curated mammalian TP53 dataset. It contains species/accession labels, branch lengths and support annotations generated from the IQ-TREE analysis.

## Audit and Reproducibility

### Project audit summary

#### TP53 Computational Evolutionary Oncology Project Audit Summary

##### Core dataset

- Mammalian TP53 sequences retained after QC: 56
- Human TP53 positions analyzed: 393
- Alignment: MAFFT
- Phylogeny: IQ-TREE / ModelFinder / Q.bird+I+G4

##### Canonical hotspot summary

**Audit table.** Extracted from supplied project audit summary.

| human position | label | domain             | structural annotation | mutation count | mutation frequency percent | hotspot rank | majority conservation | human residue conservation | shannon entropy |
|----------------|-------|--------------------|-----------------------|----------------|----------------------------|--------------|-----------------------|----------------------------|-----------------|
| 273            | R273  | DNA binding domain | DNA contact           | 268            | 6.3432                     | 1            | 1                     | 1                          | 0               |
| 248            | R248  | DNA binding domain | DNA contact           | 225            | 5.32544                    | 2            | 1                     | 1                          | 0               |
| 175            | R175  | DNA binding domain | structural core       | 167            | 3.95266                    | 3            | 1                     | 1                          | 0               |
| 245            | G245  | DNA binding domain | structural core       | 91             | 2.15385                    | 6            | 1                     | 1                          | 0               |
| 282            | R282  | DNA binding domain | structural core       | 91             | 2.15385                    | 5            | 1                     | 1                          | 0               |
| 249            | R249  | DNA binding domain | structural core       | 47             | 1.11243                    | 19           | 1                     | 1                          | 0               |

##### Structural class summary

**Audit table.** Extracted from supplied project audit summary.

| hotspot class          | structural annotation            | n positions | total mutation count | mean mutation count | mean conservation | median conservation | mean entropy |
|------------------------|----------------------------------|-------------|----------------------|---------------------|-------------------|---------------------|--------------|
| canonical hotspot      | DNA contact                      | 2           | 493                  | 246.5               | 1                 | 1                   | 0            |
| canonical hotspot      | structural core                  | 4           | 396                  | 99                  | 1                 | 1                   | 0            |
| recurrent noncanonical | DBD recurrent                    | 8           | 529                  | 66.125              | 0.997768          | 1                   | 0.0161542    |
| recurrent noncanonical | oligomerization domain recurrent | 1           | 59                   | 59                  | 0.821429          | 0.821429            | 0.676942     |

##### Hotspot vs control statistics

**Audit table.** Extracted from supplied project audit summary.

| comparison                                      | metric                     | n hotspot | n control | hotspot mean | control mean | hotspot median | control median | mean difference | mann whitney u | p value one sided hotspot greater | cliff's delta | bootstrap mean diff ci 25 | bootstrap mean diff median | bootstrap mean diff ci 975 | error | p value one sided hotspot lower |
|-------------------------------------------------|----------------------------|-----------|-----------|--------------|--------------|----------------|----------------|-----------------|----------------|-----------------------------------|---------------|---------------------------|----------------------------|----------------------------|-------|---------------------------------|
| canonical_hotspots_vs_DBD_nonhotspots           | majority_conservation      | 6         | 185       | 1            | 0.933843     | 1              | 1              | 0.0661565       | 819            | 0.0155739                         | 0.475676      | 0.0492367                 | 0.0657625                  | 0.0847739                  | nan   | nan                             |
| canonical_hotspots_vs_whole_protein_nonhotspots | majority_conservation      | 6         | 373       | 1            | 0.871531     | 1              | 0.928571       | 0.128469        | 1893           | 0.00156207                        | 0.691689      | 0.112396                  | 0.128165                   | 0.145205                   | nan   | nan                             |
| canonical_hotspots_vs_terminal_regions          | majority_conservation      | 6         | 94        | 1            | 0.783764     | 1              | 0.801786       | 0.216236        | 546            | 6.45413e-05                       | 0.93617       | 0.183076                  | 0.216261                   | 0.251186                   | nan   | nan                             |
| canonical_hotspots_vs_DBD_nonhotspots           | human_residue_conservation | 6         | 185       | 1            | 0.910195     | 1              | 1              | 0.0898052       | 819            | 0.015577                          | 0.475676      | 0.064147                  | 0.0892576                  | 0.118459                   | nan   | nan                             |
| canonical_hotspots_vs_whole_protein_nonhotspots | human_residue_conservation | 6         | 373       | 1            | 0.836497     | 1              | 0.928571       | 0.163503        | 1893           | 0.00156224                        | 0.691689      | 0.140035                  | 0.163031                   | 0.188232                   | nan   | nan                             |
| canonical_hotspots_vs_terminal_regions          | human_residue_conservation | 6         | 94        | 1            | 0.722058     | 1              | 0.785714       | 0.277942        | 546            | 6.45443e-05                       | 0.93617       | 0.224498                  | 0.277609                   | 0.334583                   | nan   | nan                             |
| canonical_hotspots_vs_DBD_nonhotspots           | shannon_entropy            | 6         | 185       | 0            | 0.303681     | 0              | 0              | -0.303681       | 291            | nan                               | -0.475676     | -0.379004                 | -0.302619                  | -0.23517                   | nan   | 0.0155877                       |
| canonical_hotspots_vs_whole_protein_nonhotspots | shannon_entropy            | 6         | 373       | 0            | 0.584684     | 0              | 0.371232       | -0.584684       | 345            | nan                               | -0.691689     | -0.651103                 | -0.583696                  | -0.520224                  | nan   | 0.00156381                      |
| canonical_hotspots_vs_terminal_regions          | shannon_entropy            | 6         | 94        | 0            | 0.973846     | 0              | 0.988118       | -0.973846       | 18             | nan                               | -0.93617      | -1.10485                  | -0.974415                  | -0.842717                  | nan   | 6.46228e-05                     |

| comparison                                      | metric             | n hot spot | n control | hot spot mean | control mean | hot spot median | control median | mean difference | mann whitney u | p value one sided hotspots greater | cliffs delta | bootstrap mean difference 25 | bootstrap mean difference median | bootstrap mean difference 975 | error | p value one sided hotspots lower |
|-------------------------------------------------|--------------------|------------|-----------|---------------|--------------|-----------------|----------------|-----------------|----------------|------------------------------------|--------------|------------------------------|----------------------------------|-------------------------------|-------|----------------------------------|
| canonical_hotspots_vs_DBD_nonhotspots           | normalized_entropy | 6          | 185       | 0             | 0.0702652    | 0               | 0              | -0.0702652      | 291            | nan                                | -0.475676    | -0.0876933                   | -0.0700194                       | -0.0544133                    | nan   | 0.0155877                        |
| canonical_hotspots_vs_whole_protein_nonhotspots | normalized_entropy | 6          | 373       | 0             | 0.135283     | 0               | 0.0858951      | -0.135283       | 345            | nan                                | -0.691689    | -0.150651                    | -0.135054                        | -0.120368                     | nan   | 0.00156381                       |
| canonical_hotspots_vs_terminal_regions          | normalized_entropy | 6          | 94        | 0             | 0.225327     | 0               | 0.228629       | -0.225327       | 18             | nan                                | -0.93617     | -0.255639                    | -0.225458                        | -0.194986                     | nan   | 6.46228e-05                      |

### Mutation-frequency conservation statistics

**Audit table.** Extracted from supplied project audit summary.

| analysis                                                 | subset                  | n   | spearman rho | p value     | n to p | n control | top mean | control mean | top median | control median | mean difference | mann whitney u |
|----------------------------------------------------------|-------------------------|-----|--------------|-------------|--------|-----------|----------|--------------|------------|----------------|-----------------|----------------|
| spearman_mutation_count_vs_majority_conservation         | all_residues            | 393 | 0.429523     | 4.48967e-19 | nan    | nan       | nan      | nan          | nan        | nan            | nan             | nan            |
| spearman_mutation_count_vs_human_residue_conservation    | all_residues            | 393 | 0.429851     | 4.19367e-19 | nan    | nan       | nan      | nan          | nan        | nan            | nan             | nan            |
| spearman_mutation_count_vs_majority_conservation         | DNA_binding_domain      | 191 | 0.367587     | 1.68499e-07 | nan    | nan       | nan      | nan          | nan        | nan            | nan             | nan            |
| spearman_mutation_count_vs_human_residue_conservation    | DNA_binding_domain      | 191 | 0.374341     | 9.56578e-08 | nan    | nan       | nan      | nan          | nan        | nan            | nan             | nan            |
| spearman_mutation_count_vs_majority_conservation         | mutated_codons_only     | 300 | 0.45583      | 8.47724e-17 | nan    | nan       | nan      | nan          | nan        | nan            | nan             | nan            |
| spearman_mutation_count_vs_human_residue_conservation    | mutated_codons_only     | 300 | 0.464684     | 1.78215e-17 | nan    | nan       | nan      | nan          | nan        | nan            | nan             | nan            |
| spearman_mutation_count_vs_majority_conservation         | DBD_mutated_codons_only | 180 | 0.411757     | 9.32279e-09 | nan    | nan       | nan      | nan          | nan        | nan            | nan             | nan            |
| spearman_mutation_count_vs_human_residue_conservation    | DBD_mutated_codons_only | 180 | 0.420555     | 4.15474e-09 | nan    | nan       | nan      | nan          | nan        | nan            | nan             | nan            |
| is_top_10_mutated_codon_vs_DNA_binding_domain_background | majority_conservation   | nan | nan          | 0.00243859  | 10     | 181       | 1        | 0.932381     | 1          | 1              | 0.0676186       | 1345           |
| is_top_20_mutated_codon_vs_DNA_binding_domain_background | majority_conservation   | nan | nan          | 0.0113359   | 20     | 172       | 0.983929 | 0.929674     | 1          | 1              | 0.0542548       | 2213           |
| is_top_30_mutated_codon_vs_DNA_binding_domain_background | majority_conservation   | nan | nan          | 0.00118198  | 30     | 163       | 0.984524 | 0.9259       | 1          | 0.982143       | 0.0586235       | 3232           |
| is_canonical_hotspot_vs_DNA_binding_domain_background    | majority_conservation   | nan | nan          | 0.0155739   | 6      | 185       | 1        | 0.933843     | 1          | 1              | 0.0661565       | 819            |

### Main interpretation

Canonical TP53 cancer hotspots are invariant across the curated mammalian TP53 alignment and show higher conservation than DNA-binding-domain background residues. Mutation recurrence in TCGA PanCancer/cBioPortal data is positively associated with mammalian residue conservation, supporting the interpretation that recurrent TP53 cancer mutations concentrate at evolutionarily constrained and functionally sensitive residues.

### Required caution

The result does not prove that conservation causes cancer recurrence. It supports an association between recurrence, functional constraint, and domain-level conservation. R249 is fully conserved but ranks lower than other canonical hotspots in this PanCancer-derived dataset, suggesting context-dependent recurrence.

### Submission readiness audit

**Audit Table A1.** Submission readiness audit.

| item                                 | path                                 | status | detail      |
|--------------------------------------|--------------------------------------|--------|-------------|
| manuscript/TP53_CBC_Manuscript_v3.md | manuscript/TP53_CBC_Manuscript_v3.md | PASS   | 39560 bytes |

| item                                                                            | path                                                                            | status   | detail              |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------|---------------------|
| manuscript/TP53_CBC_Abstract_v1.md                                              | manuscript/TP53_CBC_Abstract_v1.md                                              | PAS<br>S | 2172<br>bytes       |
| manuscript/TP53_CBC_Introduction_v1.md                                          | manuscript/TP53_CBC_Introduction_v1.md                                          | PAS<br>S | 4067<br>bytes       |
| manuscript/TP53_CBC_Results_v1.md                                               | manuscript/TP53_CBC_Results_v1.md                                               | PAS<br>S | 8757<br>bytes       |
| manuscript/TP53_CBC_Discussion_v1.md                                            | manuscript/TP53_CBC_Discussion_v1.md                                            | PAS<br>S | 6933<br>bytes       |
| manuscript/TP53_CBC_Methods_v1.md                                               | manuscript/TP53_CBC_Methods_v1.md                                               | PAS<br>S | 9142<br>bytes       |
| manuscript/TP53_CBC_Figure_Legends_v1.md                                        | manuscript/TP53_CBC_Figure_Legends_v1.md                                        | PAS<br>S | 5901<br>bytes       |
| manuscript/TP53_CBC_Back_Matter_v1.md                                           | manuscript/TP53_CBC_Back_Matter_v1.md                                           | PAS<br>S | 2429<br>bytes       |
| results/figures/Figure_Conservation_Profile_v0.png                              | results/figures/Figure_Conservation_Profile_v0.png                              | PAS<br>S | 3470<br>83<br>bytes |
| results/figures/Figure_Hotspot_vs_Control_Boxplot_v0.png                        | results/figures/Figure_Hotspot_vs_Control_Boxplot_v0.png                        | PAS<br>S | 1125<br>96<br>bytes |
| results/figures/Figure_Sensitivity_canonical_6_v0.png                           | results/figures/Figure_Sensitivity_canonical_6_v0.png                           | PAS<br>S | 1978<br>73<br>bytes |
| results/figures/Figure_Mutation_Count_vs_Conservation_v0.png                    | results/figures/Figure_Mutation_Count_vs_Conservation_v0.png                    | PAS<br>S | 2288<br>06<br>bytes |
| results/figures/Figure_TP53_Domain_Lollipop_Mutation_Map_v0.png                 | results/figures/Figure_TP53_Domain_Lollipop_Mutation_Map_v0.png                 | PAS<br>S | 2369<br>58<br>bytes |
| results/figures/Figure_TP53_Mammalian_Phylogeny_v0.png                          | results/figures/Figure_TP53_Mammalian_Phylogeny_v0.png                          | PAS<br>S | 9134<br>49<br>bytes |
| results/figures/Figure_Sensitivity_extended_14_provisional_v0.png               | results/figures/Figure_Sensitivity_extended_14_provisional_v0.png               | PAS<br>S | 1915<br>64<br>bytes |
| results/figures/Figure_Top30_Mutated_Codons_Conservation_v0.png                 | results/figures/Figure_Top30_Mutated_Codons_Conservation_v0.png                 | PAS<br>S | 2018<br>68<br>bytes |
| results/figures/Figure_TP53_Mutation_Distribution_v0.png                        | results/figures/Figure_TP53_Mutation_Distribution_v0.png                        | PAS<br>S | 2195<br>18<br>bytes |
| results/conservation/TP53_residue_conservation_v0.csv                           | results/conservation/TP53_residue_conservation_v0.csv                           | PAS<br>S | 4460<br>5<br>bytes  |
| results/conservation/TP53_conservation_with_cbiportal_mutation_frequency_v0.csv | results/conservation/TP53_conservation_with_cbiportal_mutation_frequency_v0.csv | PAS<br>S | 8771<br>8<br>bytes  |
| results/conservation/TP53_conservation_mutation_structural_annotation_v0.csv    | results/conservation/TP53_conservation_mutation_structural_annotation_v0.csv    | PAS<br>S | 9812<br>6<br>bytes  |
| results/mutation_frequency/TP53_cbiportal_mutation_frequency_by_codon_v0.csv    | results/mutation_frequency/TP53_cbiportal_mutation_frequency_by_codon_v0.csv    | PAS<br>S | 3486<br>5<br>bytes  |
| results/mutation_frequency/TP53_cbiportal_raw_mutations_v0.csv                  | results/mutation_frequency/TP53_cbiportal_raw_mutations_v0.csv                  | PAS<br>S | 4236<br>70<br>bytes |
| results/statistics/TP53_hotspot_control_statistics_v0.csv                       | results/statistics/TP53_hotspot_control_statistics_v0.csv                       | PAS<br>S | 3154<br>bytes       |
| results/statistics/TP53_sensitivity_hotspot_statistics_v0.csv                   | results/statistics/TP53_sensitivity_hotspot_statistics_v0.csv                   | PAS<br>S | 2066<br>7<br>bytes  |
| results/statistics/TP53_canonical_hotspot_summary_v0.csv                        | results/statistics/TP53_canonical_hotspot_summary_v0.csv                        | PAS<br>S | 656<br>bytes        |
| results/statistics/TP53_structural_class_summary_v0.csv                         | results/statistics/TP53_structural_class_summary_v0.csv                         | PAS<br>S | 467<br>bytes        |
| results/phylogeny/TP53_mammals_v0.treefile                                      | results/phylogeny/TP53_mammals_v0.treefile                                      | PAS<br>S | 4871<br>bytes       |
| results/phylogeny/TP53_mammals_v0.iqtree                                        | results/phylogeny/TP53_mammals_v0.iqtree                                        | PAS<br>S | 4630<br>9<br>bytes  |
| Accidental text scan                                                            | manuscript/TP53_CBC_Manuscript_v3.md                                            | PAS<br>S | none<br>found       |
| Manuscript word count                                                           | manuscript/TP53_CBC_Manuscript_v3.md                                            | PAS<br>S | 4978<br>word<br>s   |

## Sequence accession audit

**Audit Table A2.** Sequence accession audit used for reproducibility checking.

| accession    | species                    | common name                    | order          | sequence length | header                                                                          | audit flag |
|--------------|----------------------------|--------------------------------|----------------|-----------------|---------------------------------------------------------------------------------|------------|
| NP_001394193 | Homo sapiens               | Human                          | Primates       | 393             | NP_001394193 Homo_sapiens Human Primates                                        | PASS       |
| XP_001172077 | Pan troglodytes            | Chimpanzee                     | Primates       | 393             | XP_001172077 Pan_troglodytes Chimpanzee Primates                                | PASS       |
| XP_063556659 | Gorilla gorilla            | Gorilla                        | Primates       | 408             | XP_063556659 Gorilla_gorilla Gorilla Primates                                   | PASS       |
| XP_002827020 | Pongo abelii               | Orangutan                      | Primates       | 393             | XP_002827020 Pongo_abelii Orangutan Primates                                    | PASS       |
| NP_001040616 | Macaca mulatta             | Rhesus_macaque                 | Primates       | 393             | NP_001040616 Macaca_mulatta Rhesus_macaque Primates                             | PASS       |
| XP_003912321 | Papio anubis               | Baboon                         | Primates       | 393             | XP_003912321 Papio_anubis Baboon Primates                                       | PASS       |
| XP_002747994 | Callithrix jacchus         | Marmoset                       | Primates       | 395             | XP_002747994 Callithrix_jacchus Marmoset Primates                               | PASS       |
| XP_012631512 | Microcebus murinus         | Mouse_lemur                    | Primates       | 385             | XP_012631512 Microcebus_murinus Mouse_lemur Primates                            | PASS       |
| NP_001120705 | Mus musculus               | Mouse                          | Rodentia       | 381             | NP_001120705 Mus_musculus Mouse Rodentia                                        | PASS       |
| NP_001416925 | Rattus norvegicus          | Rat                            | Rodentia       | 382             | NP_001416925 Rattus_norvegicus Rat Rodentia                                     | PASS       |
| NP_001230905 | Cricetulus griseus         | Chinese_hamster                | Rodentia       | 393             | NP_001230905 Cricetulus_griseus Chinese_hamster Rodentia                        | PASS       |
| NP_001166211 | Cavia porcellus            | Guinea_pig                     | Rodentia       | 391             | NP_001166211 Cavia_porcellus Guinea_pig Rodentia                                | PASS       |
| XP_040125303 | Ictidomys tridecemlineatus | Thirteen-lined_ground_squirrel | Rodentia       | 391             | XP_040125303 Ictidomys_tridecemlineatus Thirteen-lined_ground_squirrel Rodentia | PASS       |
| NP_001297199 | Heterocephalus glaber      | Naked_mole-rat                 | Rodentia       | 391             | NP_001297199 Heterocephalus_glaber Naked_mole-rat Rodentia                      | PASS       |
| XP_005349834 | Microtus ochrogaster       | Prairie_vole                   | Rodentia       | 390             | XP_005349834 Microtus_ochrogaster Prairie_vole Rodentia                         | PASS       |
| XP_012868945 | Dipodomys ordii            | Kangaroo_rat                   | Rodentia       | 398             | XP_012868945 Dipodomys_ordii Kangaroo_rat Rodentia                              | PASS       |
| NP_001376147 | Canis lupus familiaris     | Dog                            | Carnivora      | 381             | NP_001376147 Canis_lupus_familiaris Dog Carnivora                               | PASS       |
| NP_001009294 | Felis catus                | Cat                            | Carnivora      | 386             | NP_001009294 Felis_catus Cat Carnivora                                          | PASS       |
| XP_057167334 | Ursus arctos               | Brown_bear                     | Carnivora      | 378             | XP_057167334 Ursus_arctos Brown_bear Carnivora                                  | PASS       |
| XP_004760313 | Mustela putorius furo      | Ferret                         | Carnivora      | 375             | XP_004760313 Mustela_putorius_furo Ferret Carnivora                             | PASS       |
| XP_019660679 | Ailuropoda melanoleuca     | Giant_panda                    | Carnivora      | 381             | XP_019660679 Ailuropoda_melanoleuca Giant_panda Carnivora                       | PASS       |
| XP_042822088 | Panthera tigris            | Tiger                          | Carnivora      | 386             | XP_042822088 Panthera_tigris Tiger Carnivora                                    | PASS       |
| XP_025861192 | Vulpes vulpes              | Red_fox                        | Carnivora      | 381             | XP_025861192 Vulpes_vulpes Red_fox Carnivora                                    | PASS       |
| XP_027481717 | Zalophus californianus     | California_sea_lion            | Carnivora      | 381             | XP_027481717 Zalophus_californianus California_sea_lion Carnivora               | PASS       |
| NP_776626    | Bos taurus                 | Cattle                         | Artiodactyla   | 386             | NP_776626 Bos_taurus Cattle Artiodactyla                                        | PASS       |
| NP_998989    | Sus scrofa                 | Pig                            | Artiodactyla   | 386             | NP_998989 Sus_scrofa Pig Artiodactyla                                           | PASS       |
| NP_001009403 | Ovis aries                 | Sheep                          | Artiodactyla   | 382             | NP_001009403 Ovis_aries Sheep Artiodactyla                                      | PASS       |
| XP_017920082 | Capra hircus               | Goat                           | Artiodactyla   | 382             | XP_017920082 Capra_hircus Goat Artiodactyla                                     | PASS       |
| XP_010994816 | Camelus dromedarius        | Dromedary_camel                | Artiodactyla   | 381             | XP_010994816 Camelus_dromedarius Dromedary_camel Artiodactyla                   | PASS       |
| XP_006218839 | Vicugna pacos              | Alpaca                         | Artiodactyla   | 381             | XP_006218839 Vicugna_pacos Alpaca Artiodactyla                                  | PASS       |
| XP_057572051 | Hippopotamus amphibius     | Hippopotamus                   | Artiodactyla   | 386             | XP_057572051 Hippopotamus_amphibius Hippopotamus Artiodactyla                   | PASS       |
| NP_001189334 | Equus caballus             | Horse                          | Perissodactyla | 381             | NP_001189334 Equus_caballus Horse Perissodactyla                                | PASS       |
| XP_014717043 | Equus asinus               | Donkey                         | Perissodactyla | 381             | XP_014717043 Equus_asinus Donkey Perissodactyla                                 | PASS       |
| XP_014645038 | Ceratotherium simum        | White_rhinoceros               | Perissodactyla | 397             | XP_014645038 Ceratotherium_simum White_rhinoceros Perissodactyla                | PASS       |
| XP_003416950 | Loxodonta africana         | African_elephant               | Proboscidea    | 390             | XP_003416950 Loxodonta_africana African_elephant Proboscidea                    | PASS       |
| XP_049714739 | Elephas                    | Asian_elephant                 | Proboscidea    | 390             | XP_049714739 Elephas_maximus Asian_elephant Proboscidea                         | PASS       |

| accession    | species                   | common name           | order           | sequence length | header                                                                  | audit flag |
|--------------|---------------------------|-----------------------|-----------------|-----------------|-------------------------------------------------------------------------|------------|
| 739          | maximus                   |                       |                 |                 |                                                                         | S          |
| XP_036692565 | Balaenoptera musculus     | Blue_whale            | Cetacea         | 387             | XP_036692565 Balaenoptera_musculus Blue_whale Cetacea                   | PASS       |
| XP_073653870 | Tursiops truncatus        | Bottlenose_dolphin    | Cetacea         | 393             | XP_073653870 Tursiops_truncatus Bottlenose_dolphin Cetacea              | PASS       |
| XP_033278862 | Orcinus orca              | Orca                  | Cetacea         | 400             | XP_033278862 Orcinus_orca Orca Cetacea                                  | PASS       |
| XP_028355209 | Physeter catodon          | Sperm_whale           | Cetacea         | 387             | XP_028355209 Physeter_catodon Sperm_whale Cetacea                       | PASS       |
| XP_022451328 | Delphinapterus leucas     | Beluga_whale          | Cetacea         | 387             | XP_022451328 Delphinapterus_leucas Beluga_whale Cetacea                 | PASS       |
| XP_006102640 | Myotis lucifugus          | Little_brown_bat      | Chiroptera      | 390             | XP_006102640 Myotis_lucifugus Little_brown_bat Chiroptera               | PASS       |
| XP_011357167 | Pteropus vampyrus         | Large_flying_fox      | Chiroptera      | 378             | XP_011357167 Pteropus_vampyrus Large_flying_fox Chiroptera              | PASS       |
| XP_036078506 | Rousettus aegyptiacus     | Egyptian_fruit_bat    | Chiroptera      | 383             | XP_036078506 Rousettus_aegyptiacus Egyptian_fruit_bat Chiroptera        | PASS       |
| XP_008151723 | Eptesicus fuscus          | Big_brown_bat         | Chiroptera      | 388             | XP_008151723 Eptesicus_fuscus Big_brown_bat Chiroptera                  | PASS       |
| XP_032947086 | Rhinolophus ferrumequinum | Greater_horseshoe_bat | Chiroptera      | 380             | XP_032947086 Rhinolophus_ferrumequinum Greater_horseshoe_bat Chiroptera | PASS       |
| XP_056673571 | Monodelphis domestica     | Opossum               | Didelphimorphia | 361             | XP_056673571 Monodelphis_domestica Opossum Didelphimorphia              | PASS       |
| XP_031821305 | Sarcophilus harrisii      | Tasmanian_devil       | Dasyuromorphia  | 371             | XP_031821305 Sarcophilus_harrisii Tasmanian_devil Dasyuromorphia        | PASS       |
| XP_072497274 | Macropus eugenii          | Tammar_wallaby        | Diprotodontia   | 371             | XP_072497274 Macropus_eugenii Tammar_wallaby Diprotodontia              | PASS       |
| XP_020822127 | Phascogale cinerea        | Koala                 | Diprotodontia   | 371             | XP_020822127 Phascogale_cinerea Koala Diprotodontia                     | PASS       |
| XP_060060302 | Erinaceus europaeus       | European_hedgehog     | Eulipotyphla    | 374             | XP_060060302 Erinaceus_europaeus European_hedgehog Eulipotyphla         | PASS       |
| XP_054988705 | Sorex araneus             | Common_shrew          | Eulipotyphla    | 383             | XP_054988705 Sorex_araneus Common_shrew Eulipotyphla                    | PASS       |
| NP_001075873 | Oryctolagus cuniculus     | Rabbit                | Lagomorpha      | 391             | NP_001075873 Oryctolagus_cuniculus Rabbit Lagomorpha                    | PASS       |
| XP_004594840 | Ochotona princeps         | Pika                  | Lagomorpha      | 385             | XP_004594840 Ochotona_princeps Pika Lagomorpha                          | PASS       |
| XP_023447566 | Dasypus novemcinctus      | Nine-banded_armadillo | Cingulata       | 384             | XP_023447566 Dasypus_novemcinctus Nine-banded_armadillo Cingulata       | PASS       |
| XP_023586869 | Trichechus manatus        | Manatee               | Sirenia         | 392             | XP_023586869 Trichechus_manatus Manatee Sirenia                         | PASS       |

### Permutation statistics audit

**Audit Table A3.** Permutation hotspot statistics audit.

| comparison                              | n query | observed mean conservation | null mean conservation | null 2.5 percentile | null 97.5 percentile | empirical p one sided |
|-----------------------------------------|---------|----------------------------|------------------------|---------------------|----------------------|-----------------------|
| canonical 6 vs DBD permutation          | 6       | 1                          | 0.935845368            | 0.8125              | 0.99702381           | 0.023589764           |
| top10 mutated codons vs DBD permutation | 10      | 1                          | 0.935881298            | 0.846428571         | 0.991071429          | 0.001819982           |
| top20 mutated codons vs DBD permutation | 19      | 0.992481203                | 0.935910678            | 0.876879699         | 0.979323308          | 0.001389986           |
| top30 mutated codons vs DBD permutation | 28      | 0.994260204                | 0.935896467            | 0.889587199         | 0.973133117          | 3.00E-05              |

### Software versions and computational environment

**Audit Table A4.** Software versions, packages and database resources recorded or implied by supplied files.

| Software or resource  | Recorded version or status                                        | Source note                                                      |
|-----------------------|-------------------------------------------------------------------|------------------------------------------------------------------|
| IQ-TREE               | 2.2.6 COVID-edition built Dec 8 2023                              | Recorded in TP53 mammals v0.iqtree.                              |
| ModelFinder           | Integrated in IQ-TREE analysis                                    | Best-fit model selected by BIC.                                  |
| UFBoot2               | Integrated in IQ-TREE analysis                                    | Ultrafast bootstrap analysis referenced by IQ-TREE output.       |
| MAFFT                 | Version not explicitly recorded in supplied files                 | Used for multiple sequence alignment according to project audit. |
| Python                | Version not explicitly recorded in supplied files                 | Used for computational parsing, statistics and plotting.         |
| Biopython             | Version not explicitly recorded in supplied files                 | Referenced as sequence/phylogenetic file handling dependency.    |
| NCBI Protein database | Database version/date not explicitly recorded in supplied files   | Source of TP53 protein accessions.                               |
| cBioPortal            | REST/API database snapshot not explicitly dated in supplied files | Source of TCGA PanCancer TP53 mutation-frequency data.           |



### Analysis workflow and quality-control steps

1. Assemble mammalian species list and retrieve TP53 protein sequences from NCBI Protein.
2. Apply sequence-quality checks for accession validity, length compatibility, ambiguous residues, stop codons and annotation flags.
3. Retain 56 curated mammalian TP53 protein sequences and freeze the FASTA dataset as TP53\_curated\_v0.fasta.
4. Align curated TP53 protein sequences with MAFFT and map alignment columns to 393 human TP53 residue positions.
5. Calculate residue-level conservation, entropy, gap and amino-acid diversity metrics.
6. Integrate cBioPortal/TCGA PanCancer TP53 mutation-frequency data by codon.
7. Perform domain-matched hotspot-control statistical testing and permutation analysis.
8. Perform phylogenetic sensitivity analyses and maximum-likelihood phylogenetic reconstruction with IQ-TREE.
9. Generate supplementary figures, tables, audit files and reproducibility records.

### Reproducibility statement

Another researcher can reproduce the workflow by using the supplied accession audit, curated FASTA file, residue-level conservation table, mutation-frequency table, permutation-statistics table, IQ-TREE output, treefile and audit records. The recommended reproduction route is to verify the accession list, regenerate or inspect TP53\_curated\_v0.fasta, align the curated sequences, map alignment columns to human TP53 residue numbering, recalculate conservation metrics, merge cBioPortal codon-level mutation frequencies, rerun domain-matched and permutation tests, and compare regenerated results with the supplied supplementary tables and audit files.

The supplied supplementary files preserve the key evidence required to audit sequence inclusion, residue-level conservation, mutation-frequency integration, phylogenetic model selection, permutation statistics and submission readiness.